
THE EVOLUTION OF MACHINE LEARNING AND AI

Week 2 Reference Booklet

Responsible AI, Before We Build Anything: Bias, Fairness, Transparency and Accountability as Design Constraints

Award	Certificate in Artificial Intelligence and Machine Learning Engineering (30 ECTS)
Parent programme	Higher Diploma in Science in Computer Science (NFQ Level 8)
Module	The Evolution of Machine Learning and AI (5 ECTS, Semester 1)
Week	Week 2 of 12 · Sprint 1: Fast foundations and responsible AI up front
Indicative content	Item 9 (Ethical AI: bias, fairness, accountability, transparency), threaded through every subsequent week
Learning outcome	LO6: Summarise the ethical implications of AI and machine learning, including bias, fairness and explainability
Assessed in	Mid-module quiz, Week 6 (50% of module mark); responsibility section of the final project, Week 12
Companions	Week 2 Interactive Companion (Bias & Fairness Explorer) · Week 2 live lecture · Lab: dataset and model audit, SHAP and LIME

How to read this booklet. The live lecture gives you the argument in two hours; this booklet gives you the full account, the case law and the case files. Everything in Chapters 1 to 16 is examinable in the Week 6 quiz. Boxes marked **Deeper Detail** go beyond what the lecture had time for. Boxes marked **Everyday Analogy** are the mental pictures the lecture used: they are the fastest route to remembering a concept under exam conditions. Boxes marked **Case File** are documented, sourced harms: know two of them well enough to narrate in three sentences.

Contents

1	Why Responsible AI Comes Second	4
1.1	The difference between a gate and a constraint	4
1.2	Three practical skills this week gives you	4
1.3	The one picture to keep in your head	5
1.4	How the booklet is organised	5
2	Three Senses of One Word	7
2.1	Sense one: statistical bias	7
2.2	Sense two: inductive bias	7
2.3	Sense three: societal bias	7
2.4	Two kinds of harm	8
3	Where Bias Enters: The Pipeline	9
3.1	The seven stages	9
3.2	Why the loop matters more than the stages	10
4	The World and the Data	12
4.1	The data is a photograph, and the world was not posed neatly	12
4.2	Representation bias: who is in the sample	12
4.3	Measurement bias: the instrument itself	13
4.4	Why “just collect more data” is not the fix	13
5	The Label	15
5.1	What labels actually record	15
5.2	The Amazon case, read correctly	16
5.3	Fairness through unawareness, and why it fails	16
6	The Objective and the Proxy Problem	18
6.1	Why proxies are unavoidable	18
6.2	Goodhart’s law, in production	18
6.3	The clearest case in the literature	18
7	The Model, the Average, and the Threshold	21
7.1	Aggregation: the majority writes the loss function	21
7.2	The threshold: one number, all the consequences	22
7.3	Reading the confusion matrix morally	22
8	Fairness, Formally	24
8.1	Family one: demographic parity	24
8.2	Family two: error-rate parity	24
8.3	Family three: calibration and predictive parity	24
8.4	Family four: individual and counterfactual fairness	25
9	The Impossibility Result	27
9.1	The controversy	27
9.2	The theorem	27
9.3	Why this chapter parallels Week 1’s XOR	28
10	Deployment: Oversight, Drift, and the Loop	29
10.1	Automation bias: why “human in the loop” is usually not a control	29
10.2	Distribution shift: models decay	29
10.3	The loop: when a prediction makes its own future	30
11	Case Files	31
12	Explainability, and What It Cannot Do	33
12.1	Two axes, four quadrants	33
12.2	The two post-hoc methods you will use	33

12.3	What explanation is genuinely good for	33
12.4	Four limits you must know	34
13	Accountability: Who Answers	36
13.1	The modern “computer says no”	36
13.2	Contestability: the property that most systems lack	36
14	Regulation as an Engineering Requirement	38
14.1	The AI Act’s shape: risk tiers, and use rather than technique	38
14.2	The paradox the law has just had to resolve	39
14.3	The GDPR provisions you will actually meet	39
14.4	How this lands in Ireland	40
15	The Practice	41
15.1	Two documents that are not bureaucracy	41
15.2	Three places to intervene, and what each costs	41
15.3	The metrics, written out	42
15.4	What the lab actually finds	42
15.5	The tooling	42
16	A Field Guide: Seven Questions for Any Model You Build	44
17	The Revision Core: The Whole Week on Two Pages	45
17.1	The three senses of bias	45
17.2	The seven stages, and what leaks at each	45
17.3	The four fairness families	45
17.4	The impossibility result	45
17.5	Eight one-line lessons	46
18	Glossary in Plain English	47
19	Question Bank: Practising for Week 6	50
19.1	Short answer: two to three sentences each	50
19.2	Diagnostic: name the stage, justify in two sentences	51
19.3	Concept application: one paragraph each	51
19.4	Discussion prompts: for the forum and the reading week	51
20	Further Reading and Viewing	53
	Appendix: How Week 2 Maps to the Module	54

1 Why Responsible AI Comes Second

COVERS	Orientation: why this material sits in Week 2 rather than Week 12, and what it will do to everything you build afterwards
TIME	Read first, about 15 minutes
IN BRIEF	Ethics taught last is a review gate. Ethics taught second is a design constraint. This chapter explains the difference, and what you will be able to do by the end of the booklet.

Last week ended on a thesis: scaling worked, scaling is hitting walls, and the field is turning toward science and toward lean. That is a claim about **capability**. This week is about the other axis entirely, and it is the one your employer, your regulator and the people affected by your work will actually judge you on: not what the system can do, but **what it does to people, and who answers for it**.

There is a conventional place for this material. It is the last week of the module, after everything has been built, under a heading like “ethical considerations”. Almost every AI curriculum in the world has done it that way. This module deliberately does not, and the reason is worth stating plainly, because it is the single design decision that shapes the remaining ten weeks.

1.1 The difference between a gate and a constraint

Taught at the end, responsible AI is a **review gate**. The model exists. The pipeline is built, the target variable was chosen eight months ago by someone who has since left, the training data is what could be obtained, and the deadline is Friday. In that position, “ethical considerations” can only ever be a document **about** an artefact that is already finished. It can describe. It cannot change anything, because everything that would have needed changing was decided long before anyone opened the ethics chapter.

Taught second, the same material is a **design constraint**. It arrives before you have chosen a target variable, before you have accepted a dataset, before you have accepted a default threshold, and before you have promised a stakeholder an accuracy number that will later have to be defended. In that position it changes the artefact. It changes what you collect, what you predict, what you measure, what you report, and, occasionally and most importantly, whether you build the thing at all.

KEY IDEA

The central claim of this week: by the time a model exists, most of its fairness properties have already been fixed by decisions taken before training. Those decisions were made about data, labels, target variables and thresholds, largely in meetings, largely by people who did not think of themselves as making ethical choices. Responsible AI is the discipline of **recognising those moments as they happen**, naming them, and writing them down.

1.2 Three practical skills this week gives you

First: you will locate harm in a system rather than blaming a model. By the end of this booklet, when you read that “an algorithm was biased”, you will not accept that sentence as an explanation. You will ask which of seven specific stages leaked, and you will usually be able to work it out from the reporting. That is not cynicism about journalism; it is the difference between an engineer and a spectator.

Second: you will be able to choose a fairness definition and defend the choice. This sounds like a soft skill. It is not. Chapter 9 contains a theorem, as rigorous as anything in Week 1’s XOR chapter, proving that several reasonable definitions of fairness are **mathematically incompatible**. Once you know that, “make the model fair”

stops being an instruction and becomes a question, and answering it in writing, with the domain experts present, is a professional act you will be paid for.

Third: you will read the law as a specification. The EU AI Act, the GDPR and their surrounding case law are not obstacles that arrive after the engineering. They are a requirements document with penalties attached, and they specify things you would want anyway: know your data, evaluate by subgroup, log your decisions, keep a human who can actually overrule the machine, and be able to tell an affected person why. Chapter 14 reads it that way.

1.3 The one picture to keep in your head

Week 1's picture was a wave. This week's is a **pipeline**: seven stages between the world and a decision about a person, each of which can introduce or amplify harm, and a feedback arrow at the end that turns the whole thing into a loop.

KEY IDEA

The seven stages: the world → the data → the labels → the objective → the model → the threshold → the deployment; and then the arrow back, because today's decisions become tomorrow's training data. Bias is not a substance that contaminates a model. It is a set of **choices**, distributed across those seven stages, most of which look purely technical at the moment they are made.

EVERYDAY ANALOGY

Walk through any park after a wet winter and you will see worn tracks across the grass: the **desire paths**, where people actually walk. Now imagine the council paving them. The paths were an accurate record of behaviour, and paving them is a reasonable thing to do. But something has changed: the path has stopped **describing** where people walk and started **deciding** it. Anyone who would have walked elsewhere now walks here, because paths are what you walk on.

That is exactly what a deployed model does. A faithful model of an unfair world is an **accurate** model, and putting it into production is a decision to continue the pattern it learned. The standing question for the rest of this module: **is this system describing the world, or paving it?**

1.4 How the booklet is organised

Chapters 2 to 7 walk the pipeline in order, one stage at a time: what “bias” even means (Chapter 2), the map itself (Chapter 3), then the world and the data (4), the label (5), the objective (6), and the model and threshold together (7). Chapters 8 and 9 formalise fairness and then prove that you cannot have all of it. Chapter 10 covers what happens after deployment, which is where the loop closes. Chapter 11 collects the case files. Chapter 12 is explainability and its limits, Chapter 13 accountability, Chapter 14 the law, and Chapter 15 the practice: the documents, tools and mitigations you actually use.

Chapter 16 is a field guide: seven questions for anything you ship, the counterpart to Week 1's seven questions for anything you read. Chapters 17 to 19 are your revision core: the whole week compressed to two pages, a plain-English glossary, and a bank of practice questions in the Week 6 registers. Chapter 20 is further reading, all optional.

Two reading routes are sensible. The **full route** is front to back within a week of the lecture, at roughly two hours. The **revision route**, for the days before the Week 6 quiz, is Chapter 17 first, then the Quiz Watch boxes, then the question bank in Chapter 19, dipping back only where a question exposes a gap.

QUIZ WATCH · WEEK 6

LO6 is examined in the Week 6 quiz and carries into the Week 12 project. Expect three registers: **definitional** (state and distinguish fairness criteria), **diagnostic** (given a scenario, identify which pipeline stage failed and why), and **applied** (read a confusion matrix or a fairness metric and say what it implies for a named group). The seven-stage pipeline is the scaffold for the diagnostic questions, exactly as the seaside town was for Week 1's pattern questions.

2 Three Senses of One Word

COVERS	Statistical bias, inductive bias and societal bias: why the same word means three different things, one of which is a virtue
TIME	About 12 minutes
IN BRIEF	“We removed the bias” is not a claim until you say which bias. One sense is a technical defect, one is the thing that makes learning possible at all, and one is the subject of this week.

Before anything else, a vocabulary problem that causes more confused arguments than any other in this field. The word **bias** does three separate jobs in machine learning, and people slide between them, sometimes accidentally and sometimes as a debating tactic.

2.1 Sense one: statistical bias

In statistics, an estimator is **biased** if it is systematically off in one direction: its expected value differs from the quantity it is estimating. This is the **bias** in the bias-variance trade-off you will meet formally in Week 3. A high-bias model is too simple to capture the pattern and underfits; a high-variance model chases noise and overfits. Nothing moral is at stake. A biased estimator is not unjust, merely inaccurate in a structured way.

2.2 Sense two: inductive bias

An **inductive bias** is the set of assumptions a learning algorithm brings to the problem before it sees any data, and it is the reason learning works at all. Given a finite set of examples, infinitely many functions fit them; something has to prefer one over the others, and that something is the inductive bias. A convolutional network assumes that nearby pixels are related and that a cat is a cat wherever it appears in the frame. A recurrent network assumes that order matters. A physics-informed network assumes the heat equation holds.

This sense of the word is not a defect. It is the entire subject of Weeks 4 and 5, where the module’s science-grounded strand argues that **better inductive biases** are the frontier: build in what is known and the model needs less data and generalises further. Week 5 will spend two hours deliberately adding bias in this sense and calling it a virtue.

KEY IDEA

A model with **no** inductive bias cannot learn: with no preference among the infinitely many functions consistent with the training data, there is no basis for choosing one. This is the practical content of the **no free lunch** theorems. So “our model has no bias” is either a claim about sense three, or it is nonsense.

2.3 Sense three: societal bias

The subject of this week. A system exhibits societal bias when it **systematically disadvantages a group of people**, whether or not anyone intended it, and whether or not the model is statistically well behaved. This is the only sense in which a model can **wrong** somebody, and the only one that a court, a regulator or a person refused a loan will care about.

Notice that the three senses can point in opposite directions. A model can be statistically unbiased and socially harmful, if the world it faithfully represents is unjust. A model can have strong inductive bias and be the fairest option available, if that bias is a physical law rather than a historical pattern. Confusing the senses lets people defend the third by praising the second, which is a rhetorical move you will now see everywhere.

EVERYDAY ANALOGY

Three unrelated meanings of one word, running together in a sentence, is not unusual in English: consider “a fair copy”, “a fair outcome”, and “the weather is fair”. Nobody would let an argument turn on sliding between those. Yet “unbiased model” does exactly that in vendor decks, press releases and, occasionally, papers. When you write “bias” in your project brief, put the qualifier in front of it.

2.4 Two kinds of harm

One further distinction, because the metrics in Chapter 8 only reach the first kind.

An **allocative harm** occurs when a system withholds or grants a resource or opportunity: the loan refused, the interview not offered, the parole denied, the benefit clawed back. Allocative harms are countable, litigable, and the target of essentially every fairness metric in this booklet.

A **representational harm** occurs when a system encodes or reinforces who counts as normal, without any single allocation to point at: the translation system that renders every unspecified doctor as male and every nurse as female, the image generator that produces one kind of face for “professional”, the speech recogniser that hears one accent and not another. There is no individual denial to litigate, the effects are cumulative and diffuse, and almost none of the standard metrics detect them.

KEY IDEA

Fairness metrics are overwhelmingly about **allocative** harm. That is a limitation of the tools, not a ranking of the harms. When your project brief claims a system is fair, check whether you have measured allocation and quietly said nothing about representation.

QUIZ WATCH · WEEK 6

Reliable territory: define all three senses of bias in one sentence each; explain why a model with no inductive bias cannot learn; and distinguish allocative from representational harm with one example of each. A favourite item asks you to read a sentence like “our architecture introduces no bias” and say what is wrong with it.

3 Where Bias Enters: The Pipeline

COVERS	The seven-stage map from the world to a decision about a person, and the loop that closes it
TIME	About 15 minutes; the spine of the week
IN BRIEF	Bias is not a contaminant that gets into a model. It enters at seven identifiable stages, each of which is a decision somebody made. Learn the stages and you can locate any documented harm in the news.

The most useful thing this week gives you is a **map**. When a system harms people, the reporting almost always says “the algorithm was biased”, which locates the fault inside the model, where it usually is not. The map lets you ask a better question: **which stage leaked?**

3.1 The seven stages

- 1. The world.** The first dataset is reality itself, and reality contains the accumulated results of history. No collection process can fix this, because there was nothing neutral to collect.
- 2. The data.** Somebody decided what to record, from whom, over what period, and with what instrument. Absence at this stage is invisible in an aggregate accuracy score and fatal in deployment (Chapter 4).
- 3. The labels.** Supervised learning needs a column called truth. A human made that column, using a definition, under time pressure, with their own frame of reference (Chapter 5).
- 4. The objective.** You almost never optimise the thing you care about. You optimise a **proxy** that can be measured, and the proxy has its own politics (Chapter 6).
- 5. The model.** Training minimises total loss, so a group that is five per cent of the data exerts about five per cent of the pressure. The majority writes the loss function (Chapter 7).
- 6. The threshold.** One number converts a score into a decision, and thereby allocates every error in the system between groups (Chapter 7).
- 7. The deployment.** A person or process acts on the output, in a context, under an incentive, possibly without the ability to disagree (Chapter 10).

And then the arrow back: the decisions made at stage seven generate the records that become stage two next year. The pipeline is a **loop**, which is why small initial disparities can compound rather than wash out.



Figure 3.1 · The seven stages, and what leaks at each. The arrow on the left is the reason this is a loop rather than a pipeline, and the reason small disparities compound instead of washing out.

KEY IDEA

Every stage is a **decision**, and almost every one of them looks purely technical at the moment it is made. Choosing a target variable feels like data engineering. Accepting a default threshold feels like using the library. That is precisely why harm slips in: nobody in the room experienced themselves as making an ethical choice.

3.2 Why the loop matters more than the stages

A one-pass pipeline with a small disparity produces a small injustice. A **loop** with a small disparity produces a growing one, because the system's own outputs become its evidence. Chapter 10 develops this properly, but note the shape now: if you send inspectors where the model says risk is high, you will find more violations there, those findings are logged, and next year's model is trained on a record of **where you looked** rather than **where the violations were**. The model is no longer measuring the world. It is measuring itself.

EVERYDAY ANALOGY

Point a microphone at the speaker it is feeding and you get a howl within seconds. Nothing in the microphone or the speaker is faulty; the fault is **the path from output back to input**. Feedback loops in deployed models are the same phenomenon at a slower frequency, and they are just as much a property of the wiring rather than the components.

TRY IT YOURSELF

Take any AI harm story from the last month. Before reading the analysis, decide which of the seven stages you think leaked, and why. Then read the analysis. You will be right more often than you expect, and where you are wrong, the correction is worth more than the initial guess. This is the exercise the Week 6 diagnostic questions are drawn from.

QUIZ WATCH · WEEK 6

Near-certain in some form: list the seven stages in order; given a scenario, name the stage that failed and justify it in two sentences; and explain in one sentence why the feedback arrow makes disparities compound rather than dilute.

4 The World and the Data

COVERS	Stages 1 and 2: the historical record you inherit, and who got left out of it
TIME	About 22 minutes
IN BRIEF	There is no neutral dataset because there was no neutral world to photograph. This chapter covers historical bias, representation bias, the Gender Shades audit, and why “just collect more data” is not the fix it sounds like.

4.1 The data is a photograph, and the world was not posed neatly

Your training set is a photograph of the world at a particular moment, taken by particular people with a particular instrument, for reasons of their own. Even if the collection process were flawless, the photograph would record a world in which opportunity, enforcement, wealth and medical attention have never been evenly distributed. This is **historical bias**: the data is accurate, and that is the problem.

Consider three examples that recur throughout this booklet:

- Criminal justice data records **arrests**, not offences. Where policing is more intense, more offences are detected, so the record encodes enforcement patterns as if they were behaviour.
- Hiring data records **who was hired and retained**, not who could have done the job. Everyone who was never interviewed is absent from the outcome column entirely.
- Health data records **who presented at a hospital and what was done for them**, not who was ill. People who cannot easily reach care appear healthier.

KEY IDEA

The engineering question is never “is this data biased?” (the answer is always yes, in some sense) but “**biased in what specific way, by what process, and does that matter for this particular decision?**” Historical bias in a dataset of star positions is irrelevant to your model. Historical bias in arrest records is the entire ballgame if you are predicting reoffence.

4.2 Representation bias: who is in the sample

The second and more tractable problem is **representation bias**: some groups are present in far smaller numbers than others, or under systematically different conditions. Because training minimises total loss, a model has little incentive to do well on a small group, and because evaluation usually reports a single average, nobody notices.

■ CASE FILE: Gender Shades, 2018

Joy Buolamwini and Timnit Gebru audited commercial gender-classification products from major vendors, and did something the field had largely not done: they **reported accuracy by subgroup** rather than in aggregate.

The results were stark. Error rates for lighter-skinned men were under one per cent. Error rates for darker-skinned women reached **34.7 per cent** in the worst case. The same product, the same “accuracy”, and a difference of more than an order of magnitude depending on who was in front of the camera.

Two details matter as much as the headline. First, **why nobody had noticed**: the standard face benchmarks of the period were themselves overwhelmingly lighter-skinned and male, so the exam shared an assumption with the syllabus and could not detect it. Second, **what the authors actually contributed**: not the criticism but the instrument. They constructed a deliberately balanced benchmark and published disaggregated numbers, which is

what made the claim impossible to argue with. That is Week 1's benchmark-culture lesson arriving exactly on schedule: claims ride numbers on shared tests that rivals can run.

EVERYDAY ANALOGY

A tailor who measures only the mannequins in the shop window will cut a beautiful suit, and will be genuinely puzzled when most customers complain about the fit. From inside the shop, the problem appears to be the customers. The measurements were accurate. The **sample** was the fiction.

KEY IDEA

Report accuracy per slice, or do not report accuracy. Disaggregated evaluation is the cheapest responsible-AI practice in this module: it requires no new algorithm, no new data, and about twenty lines of code. It is also the one most often skipped, because an aggregate number is flattering and a table of subgroup numbers rarely is.

4.3 Measurement bias: the instrument itself

A third, subtler entry point. **Measurement bias** arises when the way a feature is recorded differs systematically between groups. A wearable that measures heart rate optically performs differently on different skin tones. A credit feature such as “length of credit history” measures access to credit as much as reliability. A school-quality feature measures the school's resourcing as much as the pupil's ability. In each case the number is real and the comparison across groups is not like for like.

DEEPER DETAIL: Selection and survivorship: the labels you never see

The most dangerous form of representation bias is the one where the missing data is **caused by the decision you are automating**. Consider a lender building a repayment model. It has repayment outcomes only for applicants it **approved**. The people it refused have no outcome, because they were never given the loan. This is the **selective labels** problem, and it has a nasty consequence: the model can only learn about the region of the space where the old policy said yes, and its apparent excellent performance is partly a measure of how well it agrees with that old policy.

The same shape appears in medicine (you only see outcomes for patients who were treated), in recruitment (you only see performance for people who were hired), and in policing (you only observe crime where you patrolled). The statistical machinery for handling it exists and is not simple: it borrows from causal inference and from the economics of programme evaluation, and often the only clean solution is a deliberate randomised exploration, which raises its own ethical questions. Recognising the problem is a Week 2 skill; solving it is a research area. For the purposes of your project, the requirement is to **notice and state** when your outcome column exists only for people a previous decision selected.

4.4 Why “just collect more data” is not the fix

The intuitive remedy for representation bias is more data. It is usually wrong in one of two ways.

More data **of the same kind** reproduces the same skew, at higher confidence. If your collection process systematically under-samples a group, scaling it up under-samples them more, with tighter error bars. What is needed is **targeted** collection, which means going deliberately to the people who are missing.

And that raises the second problem, which is consent. Several widely used face datasets were assembled by scraping images of people who never agreed to be in them, and were later withdrawn amid legal and ethical

objection. The people most likely to be under-represented in a dataset are frequently the people with the strongest reasons to distrust being enrolled in one. Under the EU AI Act, **untargeted scraping of facial images from the internet or CCTV to build recognition databases is now a prohibited practice.**

KEY IDEA

Improving representation and respecting consent **pull against each other**. This is a genuine tension, not a puzzle with a clean answer. Professional practice is to state the tension explicitly in your datasheet, describe how you resolved it, and record what your dataset therefore does **not** cover, rather than implying you achieved both.

✓ IN PRACTICE: Stage 1 and 2 checklist

- Write down **who collected this data, when, and for what original purpose**. Data reused outside its original purpose is where most surprises live.
- Produce a **population table**: counts by every group you can lawfully identify, compared against the population the system will actually serve. A model trained on national data and deployed in one city has a representation problem even if the national sample was perfect.
- Evaluate **by subgroup as a matter of routine**, including intersections. Gender Shades' finding lived at the **intersection** of skin tone and sex; either variable alone would have understated it.
- Record what the dataset does **not** cover, and the tasks for which it must not be used.

QUIZ WATCH · WEEK 6

Be able to: distinguish historical, representation and measurement bias with one example each; state the Gender Shades finding, including the best and worst subgroup error rates and why the benchmark had hidden it; and explain in two sentences why “collect more data” can fail as a remedy.

5 The Label

COVERS	Stage 3: ground truth is a human judgement wearing a lab coat
TIME	About 18 minutes
IN BRIEF	Every supervised model has a column called truth, and somebody made it. This chapter covers label bias, annotator effects, proxy labels, and the Amazon recruiting case, which is really a case about proxies rather than about a word list.

Supervised learning requires a column called **the truth**. The name is the problem. In the physical sciences the label is often a genuine measurement: the mass of the sample, the temperature of the reactor. In the applications where AI meets people, the label is almost always a **human judgement**, recorded by a person or a process with a definition, a time budget and a point of view.

5.1 What labels actually record

Look closely at three labels that appear constantly in this field:

“Reoffended” almost never means “committed another offence”. It means **re-arrested**, or **reconvicted**, within some window. Those are records of the criminal justice system’s activity, and that activity is not uniform across neighbourhoods. A model that predicts this label well is predicting **police and court behaviour** with the accuracy we attribute to predicting crime.

“Good hire” usually means something like “was hired, stayed two years, and received a favourable review”. Every term in that definition passes through the preferences of the managers who did the hiring and the reviewing. If those preferences favoured a particular background, the label encodes it as ground truth.

“Toxic comment” means “annotators marked it toxic”. Research on comment-moderation datasets has repeatedly found that text written in some dialects, notably African-American English, is rated as more offensive by annotators unfamiliar with it, so models trained on those labels flag ordinary speech from those communities at elevated rates.

EVERYDAY ANALOGY

Teaching to the answer key is a perfectly good strategy, right up to the moment you notice that one examiner wrote every key, in one building, in 2009, and marked down anything phrased differently from how they would have phrased it. The student who scores perfectly has learned the examiner, not the subject.

DEEPER DETAIL: Annotator effects, and why disagreement is data

Labelling is usually done by multiple annotators and reconciled by majority vote. That reconciliation **destroys information**. When annotators disagree systematically, the disagreement is often the most interesting thing in the dataset: it marks cases where “truth” is contested, and the direction of disagreement frequently correlates with the annotators’ own backgrounds.

Better practice, increasingly common in research datasets, is to retain the full annotation distribution rather than the majority label, to report inter-annotator agreement **by annotator group**, and to treat high-disagreement items as a distinct category rather than noise to be resolved. For your own projects the practical version is simpler: record who labelled your data and how, and if a single person or a single vendor labelled everything, say so, because that fact bounds what any fairness claim about the model can mean.

5.2 The Amazon case, read correctly

■ CASE FILE: Amazon's experimental recruiting engine, reported 2018

Amazon built an internal experimental tool to score job applicants from one to five stars, trained on roughly a decade of CVs the company had received. The label it learned to reproduce was, in effect, **who we hired before**, in a sector whose technical intake had been overwhelmingly male.

According to reporting, the system learned to penalise CVs containing the word “women’s” (as in “women’s chess club captain”) and to downgrade graduates of two all-women’s colleges. Amazon edited the model to make it neutral to those particular terms. It could not satisfy itself that the model would not simply find **other** routes to the same result, and the project was abandoned.

The lesson is not “they used a bad word list.” It is the final sentence. Removing the explicit signals did not remove the information, because the data is dense with **proxies** for it: the sports played, the phrasing of achievements, the schools, the gaps in employment, the vocabulary. A capable learner will find them, because finding predictive structure is precisely what it was built to do.

5.3 Fairness through unawareness, and why it fails

This brings us to the most common mistake made in good faith by competent engineers.

The claim: if we do not give the model the protected attribute, it cannot discriminate on it.

The reality: postcode carries ethnicity and income in most countries. Height and weight carry sex. Employment gaps carry parenthood and disability. Device type and browser carry income. First names carry ethnicity, generation and often religion. Shopping history carries pregnancy. A model with a few hundred features and a few thousand examples can reconstruct a protected attribute to high accuracy without ever being shown it, and it will do so if the attribute is predictive of the label.

KEY IDEA

Fairness through unawareness blinds the model to the attribute but not to the data, and it blinds **you** to the outcome. Once you have refused to record the attribute, you cannot compute a single fairness metric in Chapter 8, because every one of them requires knowing who is in which group. You have not removed the disparity. You have removed your ability to see it. This is the paradox that Chapter 14 shows the law itself has just had to confront.

TRY IT YOURSELF

Take any tabular dataset with a protected attribute and try to **predict that attribute from the other columns**. Report the accuracy. In most real datasets it will be far higher than people expect, and the exercise takes ten minutes. It is the single most persuasive demonstration you can run for a stakeholder who believes that dropping a column solved the problem, and it makes an excellent paragraph in your final project.

✓ IN PRACTICE: Stage 3 checklist

- Write one sentence beginning **“our label actually records...”** and make it honest. If the honest version differs from the name of the column, that gap is a risk you must state.
- Record **who** produced the labels, under what instructions, and with what agreement rate.
- Where the label is a proxy for something unobservable, name the unobservable thing and the ways the proxy diverges from it.
- Never claim neutrality on the basis of having dropped a protected column.

QUIZ WATCH · WEEK 6

High-probability items: explain what the label “reoffended” actually records and why that matters; state the Amazon case in three sentences **and** give the correct lesson from it; and explain why fairness through unawareness fails, naming at least two proxy variables.

6 The Objective and the Proxy Problem

COVERS	Stage 4: the target variable, the most consequential decision in most projects
TIME	About 20 minutes
IN BRIEF	You almost never optimise the thing you care about; you optimise something measurable that stands in for it. This chapter covers proxy targets, Goodhart's law, and the Obermeyer study, the cleanest documented example of a harm caused by a target variable.

Ask an engineer what their model does and they will describe the architecture. Ask what it **optimises** and you get the loss function. Ask what the loss function is computed against and you arrive, eventually, at a column in a table, chosen months earlier, in a meeting, by people deciding what could realistically be measured. That choice is the **target variable**, and in most applied projects it is the single decision with the largest fairness consequence.

6.1 Why proxies are unavoidable

The things institutions care about are rarely measurable. “Health need”, “job performance”, “creditworthiness”, “student potential”, “risk to the public” are all constructs: real, consequential, and not present as a column in any database. So we substitute something we can count: future costs, retention, repayment, exam results, re-arrest.

This substitution is not a mistake. It is **necessary**, and doing it well is a large part of applied machine learning. The mistake is forgetting that it happened.

EVERYDAY ANALOGY

You wanted the house to be warm, so you installed the thermostat. It went in the hallway, because that is where the wiring was. Now the hallway is reliably twenty degrees, the north bedroom is freezing, and the boiler is behaving **exactly as designed**. Nobody made an error. The system is optimising a proxy for the thing you wanted, and the gap between the proxy and the goal is where all the cold people live.

6.2 Goodhart's law, in production

The economist Charles Goodhart's observation, in its popular form: **when a measure becomes a target, it ceases to be a good measure**. Machine learning is a Goodhart engine. Its entire purpose is to push hard on a measure, so any divergence between the proxy and the goal is not merely inherited but **amplified**: the optimiser will find and exploit exactly the region where the proxy and the goal come apart, because that is where the cheap wins are.

KEY IDEA

Two questions to ask of any target variable, before training. **First**: what do we actually care about, stated in plain language, with no reference to available data? **Second**: in what specific circumstances does our measurable proxy diverge from that, and does that divergence fall unevenly on identifiable groups? If the answer to the second is yes, you have found your project's principal fairness risk, and you have found it early enough to do something about it.

6.3 The clearest case in the literature

► **CASE FILE: A health algorithm and its target variable** • Obermeyer and colleagues, *Science*, 2019

A commercial risk-prediction system, applied to a very large number of patients in the United States, was used by health systems to identify people who should be enrolled in additional care-management programmes: extra nursing time, closer follow-up, coordination of chronic conditions.

Its target variable was **future healthcare costs**. This is a defensible choice: costs are recorded reliably, they are available at scale, and they correlate strongly with illness. Sicker people cost more.

The researchers found that at **the same predicted risk score**, Black patients were substantially **sicker** than white patients, measured by number and severity of chronic conditions. The system was therefore systematically under-enrolling patients who needed care most.

The mechanism was not malice and involved no protected attribute: race was not an input. Less money is spent on Black patients at equal levels of illness, for reasons of access, insurance, distance, and unequal treatment within the system. The model learned **spending** accurately, and spending had inherited the inequity of the system that produced it.

When the researchers changed the target from cost to direct measures of health, the proportion of Black patients identified for additional help rose from **17.7 per cent to 46.5 per cent**.

Read that case three times, because it dismantles three comfortable beliefs at once.

- 1. No protected attribute was used.** Blinding the model achieved nothing, exactly as Chapter 5 predicted.
- 2. No code was wrong.** The pipeline was competent, the training was sound, the reported accuracy was genuine. An audit of the model would have found nothing.
- 3. The fault was one design decision**, taken before any model existed, in a room where nobody thought they were making an ethical judgement. They were choosing a column.

KEY IDEA

The most consequential fairness decision in a typical project is made in a meeting about **what to predict**, months before anyone trains anything, and it is usually made by people who would not describe themselves as working on AI ethics. This is the strongest single argument for teaching this material in Week 2 rather than Week 12: by Week 12 the column has been chosen.

DEEPER DETAIL: Proxy risk beyond fairness: reward hacking and the alignment lineage

The proxy problem is not confined to fairness, and recognising its wider form will help you in Weeks 7 and 10. Any system optimising a measurable stand-in for an unmeasurable goal is subject to **specification gaming**: recommender systems optimising watch time and discovering that outrage retains attention; content systems optimising engagement and learning that extremity engages; reinforcement-learning agents that find a scoring bug rather than playing the game.

Large language models tuned with reinforcement learning from human feedback (Week 7) are trained against a **learned model of human preference**, which is a proxy for human judgement, which is itself a proxy for the thing wanted. Sycophancy, in which a model agrees with the user because agreement was rated highly, is a textbook proxy failure. When Week 7 introduces RLHF as the step that turned a text predictor into an assistant, ask the Week 2 question: **whose preferences, collected how, standing in for what?**

✓ IN PRACTICE: Stage 4 checklist

- Write the **goal** and the **target variable** as two separate sentences and put them side by side in the model card. If they are identical, you are either in a rare well-posed problem or you have not looked closely enough.
- List the mechanisms by which the proxy diverges from the goal, and ask, for each, whether it falls unevenly on a group.

- Where feasible, **validate against an outcome you trust more**, even on a small sample. Obermeyer's finding came from comparing the model's score against direct clinical measures on the same patients.
- Prefer changing the target over patching the output. It is nearly always cheaper, more defensible, and more effective.

QUIZ WATCH · WEEK 6

Very likely: state the Obermeyer case in three sentences including both percentages; explain why using race as an input was not necessary for the harm to occur; and define Goodhart's law and say why machine learning amplifies it.

7 The Model, the Average, and the Threshold

COVERS	Stages 5 and 6: aggregation effects in training, and the single number that allocates every error
TIME	About 20 minutes
IN BRIEF	Training optimises a total, so minorities exert minority pressure on the loss. Then one threshold turns scores into decisions and distributes the errors. This chapter is the theory behind the Week 2 lab.

7.1 Aggregation: the majority writes the loss function

Training minimises loss summed over the training set. A group that constitutes five per cent of the data therefore contributes about five per cent of the gradient signal. If serving that group well requires a different decision boundary from the one that serves the majority, the optimiser will trade their errors away, and it will be **right to do so** by the objective you gave it.

This is **aggregation bias**: a single model applied to groups for which the relationship between features and outcome genuinely differs. The classic medical example is that HbA1c thresholds for diagnosing and managing diabetes do not carry identical meaning across ethnic groups; a single model fitted to a pooled population encodes a compromise that may fit no group well.

EVERYDAY ANALOGY

The average shoe fits nobody exactly, and it fits some people extremely badly. Averages are not neutral summaries; they are decisions about whose deviation counts as noise.

The remedies all cost something, and the cost should be stated rather than hidden:

- **Reweighting** the loss to give under-represented groups more influence. Simple, effective, and it lowers average accuracy.
- **Separate models per group**, which fits each better but requires using the protected attribute explicitly at inference, with legal consequences in the EU, and fragments your data.
- **Worst-group objectives** (group distributionally robust optimisation), which optimise the worst-performing group rather than the mean. Principled, and it can be noticeably harder to train stably.

KEY IDEA

There is nearly always a trade between **average** performance and **worst-group** performance. That trade is a value judgement dressed as a hyperparameter. Report both numbers. A model card that reports only the average has concealed the decision, whether or not anyone intended to.

DEEPER DETAIL: The lean turn has a fairness bill: a warning for Week 9

Week 9 covers making models lean: pruning, low-bit quantization, and distillation. Compression does not degrade a model evenly across its inputs. A consistent research finding is that compressed models lose most of their accuracy on a comparatively small set of **atypical, long-tail** examples, while headline accuracy barely moves. Those long-tail examples are disproportionately drawn from under-represented groups, precisely because under-representation is what makes examples atypical to the model.

The practical consequence is uncomfortable and rarely stated in efficiency papers: a compression step reported as “0.4 per cent accuracy loss” may carry a much larger loss for a specific subgroup, invisible in the aggregate.

When you evaluate a lean model in Week 9, or in your final project, evaluate it **by subgroup before and after compression**, and report the deltas separately. This is one of the clearest places in the module where the responsible-AI strand and the lean strand collide, and it is an excellent final-project topic.

7.2 The threshold: one number, all the consequences

Most classifiers output a score. A decision requires a **cut**. That cut is not learned; it is chosen, and it is where the abstract disparity of the previous five stages becomes concrete allocation.

Picture two groups whose score distributions overlap but whose centres differ, for all the reasons in Chapters 4 to 6. Apply one identical rule to everybody, say accept above 0.5. This is the very definition of equal treatment: the same rule, applied without regard to group. And it produces markedly different selection rates, because the distributions it is applied to differ. Nothing in the code is unequal. The world in the data was.

Now try to fix it by equalising the selection rates. To do that with a single model you must apply **different thresholds to different groups**, which means using the protected attribute explicitly at decision time. That is ethically arguable, statistically straightforward, and in the EU it edges directly into **direct discrimination**, which is far harder to justify than the indirect kind. There is no neutral option on this slide. There is only a documented choice and an undocumented one.

KEY IDEA

Not choosing a threshold is choosing 0.5. The default in your library is a policy: adopted silently, by nobody, in a codebase with your name on the commit. The Week 2 lab exists to make this visceral: you will move a slider and watch four fairness metrics move in different directions, and discover that you cannot satisfy them all.

EVERYDAY ANALOGY

The height bar at a fairground ride. Move it two centimetres and a different set of children ride, and a different set cry. The bar is not a fact about safety; it is a decision about who bears the risk and who bears the disappointment, painted on a post so it looks like a measurement.

7.3 Reading the confusion matrix morally

The four cells of a confusion matrix are usually read as arithmetic. Every rate in Chapter 8 is built from them:

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad \text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \quad \text{PPV} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Now read the cells as **people**.

	Predicted YES	Predicted NO
Actually YES	True positive. Correctly identified.	False negative. The illness missed, the fraud undetected, the qualified applicant passed over.
Actually NO	False positive. The healthy patient sent for biopsy, the innocent family investigated, the honest claim refused.	True negative. Correctly cleared.

The two error cells are not the same size in human terms, and they do not fall on the same people. In cancer screening a false positive costs anxiety and an invasive test; a false negative can cost a life. In benefit-fraud

detection a false positive costs a family months of investigation and demanded repayments; a false negative costs the exchequer a small sum. Any threshold trades one error against the other, and **the trade lands on different groups when the score distributions differ.**

KEY IDEA

Fairness, operationally, is **which group gets which error**. Before tuning anything, answer in writing: who bears a false positive, who bears a false negative, and are they the same people? If the answer is “different people, unequally”, you have stopped doing statistics and started allocating harm, and you owe somebody a reason.

TRY IT YOURSELF

In the Bias & Fairness Explorer, set the threshold to maximise overall accuracy and record all four metrics. Then set it to equalise selection rates, then to equalise false-positive rates. Record the accuracy cost of each, and write two sentences on which you would defend for (a) a cancer screen and (b) a fraud flag. Bring both answers to the lab; they are the seed of a Chapter 8 discussion and a strong forum post.

QUIZ WATCH · WEEK 6

Reliable: explain aggregation bias in two sentences; explain why an identical threshold applied to two groups can produce very different selection rates; state which error a screening system should prefer and why; and give one reason per-group thresholds are legally risky in the EU.

8 Fairness, Formally

COVERS	The four families of fairness definition, what each one protects, and what each one costs
TIME	About 22 minutes
IN BRIEF	“Make it fair” is not an instruction until you say which fairness. Four families: demographic parity, error-rate parity, calibration, and individual or counterfactual fairness.

Everything so far has been diagnostic. This chapter makes it measurable. There are dozens of published fairness criteria; they cluster into four families, and knowing the families is enough to have a professional conversation.

Throughout, let A be a protected attribute (group membership), Y the true outcome, and $\hat{Y}$ the model’s decision.

8.1 Family one: demographic parity

The requirement: the selection rate is the same for every group. Formally, $\hat{Y}$ is independent of A :

$$P(\hat{Y} = 1 \mid A = a) = P(\hat{Y} = 1 \mid A = b) \quad \forall a, b$$

What it protects: equality of outcome. If your account of the harm is that a group has historically been under-selected and the point of the system is to change that, this is the criterion that expresses it.

What it costs: it ignores whether the groups differ on Y . If the base rates genuinely differ, enforcing equal selection rates forces the model to select less-qualified members of one group or reject better-qualified members of another by the model’s own lights, which is precisely the objection its critics raise. It also says nothing about

errors: a system could satisfy demographic parity while getting one group’s decisions right and the other group’s essentially at random.

8.2 Family two: error-rate parity

The requirement: equal error rates across groups. **Equalised odds** demands equal true-positive rates and equal false-positive rates:

$$P(\hat{Y} = 1 \mid Y = y, A = a) = P(\hat{Y} = 1 \mid Y = y, A = b) \quad \forall a, b, \quad y \in \{0, 1\}$$

Equal opportunity is the weaker version, requiring the condition only at $y = 1$, so that qualified people have the same chance of being recognised regardless of group.

What it protects: equality of **treatment by the classifier**. A false accusation should not be more likely because of who you are. This is usually the most defensible criterion where the decision is **punitive**: fraud, policing, exam invalidation, investigation.

What it costs: it takes the label Y as legitimate. If your label encodes enforcement rather than behaviour (Chapter 5), equalising errors with respect to it inherits that problem. It is also the family that collides hardest with calibration, as Chapter 9 shows.

8.3 Family three: calibration and predictive parity

The requirement: a score means the same thing for everybody. Writing S for the model’s score:

$$P(Y = 1 \mid S = s, A = a) = s \quad \forall a, s$$

Among all people given a score of 0.7, about 70 per cent should have the positive outcome, **within each group**.

Predictive parity is the thresholded version: equal positive predictive value, $P(Y = 1 \mid \hat{Y} = 1, A = a)$.

What it protects: the **meaning** of the number. A decision-maker who cannot rely on a score meaning the same thing for different people cannot rationally use it at all, which is why vendors of risk scores treat calibration as the fundamental requirement.

What it costs: when base rates differ, calibration **forces** unequal error rates. This is not a design failure. It is a theorem, and it is Chapter 9.

8.4 Family four: individual and counterfactual fairness

Individual fairness, proposed by Dwork and colleagues in 2012: similar individuals should receive similar outcomes. Given a similarity metric d on individuals and a Lipschitz constant L :

$$d(x, x') \leq \varepsilon \implies |f(x) - f(x')| \leq L \cdot d(x, x')$$

Appealing, and it moves the difficulty rather than removing it, because it requires a **similarity metric** saying which differences between two people are relevant to this decision. Constructing that metric is the entire moral question in disguise.

Counterfactual fairness, proposed by Kusner and colleagues in 2017: a decision is fair if it would have been the same in a counterfactual world where the individual belonged to a different group, holding everything not caused by group membership constant. Conceptually the sharpest of the four, and it requires a **causal model** of how the protected attribute influences everything else. You will rarely have one, and disagreements about the causal graph are disagreements about society.

KEY IDEA

Each family answers a **different question**: parity asks about outcomes, error-rate parity asks about mistakes, calibration asks about meaning, individual fairness asks about the person. None is universally correct. Choosing between them is a judgement about **what the harm is**, which is a question about the world and not about the code, and it must be made with domain experts and, where possible, affected people in the room.

DEEPER DETAIL: Why the group has to be defined, and by whom

Every group criterion presupposes that groups exist and are known. In practice this is contested at every point. Which attribute: sex, gender, both? Which categories: the census's, the client's, the person's own? What about people at intersections, where Gender Shades showed the largest effects but where sample sizes are smallest? What about attributes that are not recorded because recording them is itself intrusive, such as disability or sexual orientation?

There is no purely technical answer. Sensible practice: use the categories the **relevant law and the affected community** use rather than ones invented for convenience; test at intersections wherever counts allow; report the counts alongside the metrics so a reader can see where the estimates are unreliable; and treat inferred group membership with great care, since inferring a protected attribute in order to measure fairness is itself a processing operation requiring a legal basis (Chapter 14).

✓ IN PRACTICE: Choosing a criterion, defensibly

- **Start from the harm, not the metric.** Write one sentence describing what going wrong looks like for a person. The sentence usually points at a family.
- **If the decision is punitive**, equalising false-positive rates is usually the defensible commitment.
- **If the label is suspect**, be reluctant to calibrate to it: you would be laundering the label's problem into an authoritative number.

- **If the base-rate difference is itself the injustice you are trying to address**, demographic parity is arguable, but say so explicitly, because it is the criterion most likely to be legally challenged.
- **Write the choice down before training**, with the reasoning and the names of the people who agreed it. This document is what “accountability” concretely means.

QUIZ WATCH · WEEK 6

Core territory: define all four families in one sentence each; give one context where each is the appropriate choice; and explain why equal opportunity is weaker than equalised odds. A common applied item gives you a small confusion matrix per group and asks which criteria are satisfied.

9 The Impossibility Result

COVERS	The COMPAS controversy, and the theorem that ended it: why several reasonable fairness criteria cannot hold at once
TIME	About 20 minutes; this chapter is Week 2's XOR
IN BRIEF	Two parties argued about a risk score. Both were right, because they were measuring incompatible things. The incompatibility is provable, and unlike XOR it has no escape route.

9.1 The controversy

► CASE FILE: COMPAS · ProPublica, 2016

COMPAS is a commercial risk-assessment instrument used in parts of the United States to inform decisions about bail, sentencing and parole. In 2016 the news organisation ProPublica obtained risk scores for several thousand defendants in one Florida county, together with what actually happened over the following two years, and analysed the errors by race.

Their headline findings. Among defendants who did **not** go on to reoffend, Black defendants were roughly **twice as likely** as white defendants to have been labelled higher risk (about 45 per cent against about 23 per cent): unequal **false positives**. And among those who **did** reoffend, white defendants were considerably more likely to have been labelled lower risk (about 48 per cent against about 28 per cent): unequal **false negatives**.

The vendor's response was not a denial of these numbers. It was a different measurement: the instrument was **calibrated**. Among defendants given a particular score, the observed reoffence rate was approximately the same for both groups, so the score meant what it said regardless of race.

Independent researchers verified both claims. **Both parties were right about their own numbers**, and the disagreement could not be resolved by looking at the data more carefully.

9.2 The theorem

Work by Kleinberg, Mullainathan and Raghavan, and independently by Chouldechova, established the reason, and it is a result about arithmetic rather than about COMPAS.

KEY IDEA

The impossibility result. If two groups have different base rates of the outcome, and the classifier is not perfect, then it is impossible for a risk score to simultaneously satisfy: **calibration** (a score means the same thing in each group), **equal false-positive rates**, and **equal false-negative rates**. Any two can be achieved. All three cannot. The only escapes are **equal base rates** or a **perfect classifier**, neither of which occurs in the applications where this matters.

The intuition is worth carrying even without the algebra. Calibration ties the score to the observed frequency of the outcome within each group. If the outcome is genuinely more frequent in one group, then a well-calibrated score must, on average, be higher in that group. But a higher average score against a fixed threshold produces more positive predictions, and among the people who will not experience the outcome, more positives means more **false positives**. The error rates are forced apart by the requirement that the number mean the same thing.

9.3 Why this chapter parallels Week 1's XOR

The structural resemblance to the XOR episode is worth making explicit, because it is also the structural **difference** that matters most.

XOR, 1969	Impossibility, 2016 to 2017
A correct proof about what a single-layer perceptron cannot do.	A correct proof about what no classifier whatsoever can do.
Reached only the simplest version of the idea; multi-layer networks were unaffected.	Reaches every model, every dataset, every architecture, at any scale.
There was an escape route: a hidden layer. It took seventeen years to learn to train one.	There is no escape route. No amount of data, compute or cleverness dissolves it.
Consequence: the field over-read the result and froze a research programme for a decade.	Consequence: the field must stop asking whether a model is fair and start asking which fairness, and why .

Week 1's discipline for reading criticism was: **which version of the idea does this reach, and is there an escape route?** Apply it honestly here and the answer is uncomfortable: this criticism reaches everything, and there is no escape route. That is rare, and it is why this result is worth knowing precisely rather than approximately.

KEY IDEA

The theorem does not end the engineering work. It **relocates** it. "Is this model fair?" is not a well-posed question. "Which fairness criterion does this decision require, on what account of the harm, at what measured cost to the others, agreed by whom?" is well posed, answerable, and documentable. The theorem's real gift is that it makes the second question unavoidable.

DEEPER DETAIL: Where you can still make progress

It would be easy to read this chapter as counselling despair. It should not be. Three things remain fully available.

First, reduce the error rate. The trade-offs are forced by **imperfection**; a better model shrinks the region in which they bind. Accuracy improvements are fairness improvements, up to a point, and this is the least discussed implication of the theorem.

Second, question the base rates. The theorem takes the base rates as given. But base rates are measured with your label (Chapter 5), and if the label encodes enforcement rather than behaviour, then the "difference in base rates" that forces the trade-off may itself be an artefact of stage three. Fixing the label can dissolve a trade-off that looked mathematically compulsory.

Third, change the decision. The theorem constrains **binary classification against a fixed threshold**. Many real decisions need not take that shape: you can return a score with an interval, route uncertain cases to a human with time to consider them, offer a route of appeal, or intervene supportively rather than punitively, in which case a false positive costs somebody an offer of help rather than an investigation. Several of the most effective real-world responses to this literature have been redesigns of the **decision**, not the model.

QUIZ WATCH · WEEK 6

Near-certain in some form. Be able to: state the three criteria that cannot coexist and the two conditions under which they can; explain in plain language why calibration forces unequal error rates when base rates differ; narrate the COMPAS dispute in three sentences showing that both parties were correct; and contrast this result with XOR on the specific point of whether an escape route exists.

10 Deployment: Oversight, Drift, and the Loop

COVERS	Stage 7 and the arrow back: automation bias, distribution shift, and feedback loops
TIME	About 18 minutes
IN BRIEF	The model is now in a room with people, incentives and deadlines. This chapter covers why “a human reviews it” is usually not a control, why models decay, and why outputs that become inputs make small disparities grow.

10.1 Automation bias: why “human in the loop” is usually not a control

The most common safeguard in industry is a sentence: **a human reviews every decision**. It is also the least examined, because the psychology works against it.

Automation bias is the well-documented tendency of people to over-trust automated output, both by accepting incorrect recommendations (errors of commission) and by failing to act when the automation says nothing (errors of omission). It intensifies under exactly the conditions in which these systems are deployed: high caseloads, throughput targets, time pressure, and reviewers who are less expert than the system is presented as being. Agreeing with the machine is fast and defensible. Disagreeing requires evidence, time, and a willingness to be the person who overruled the model on the file that later went wrong.

EVERYDAY ANALOGY

Every year, drivers follow satnav instructions into rivers, up staircases and down farm tracks. Not one of them believed the road was a river. They believed the machine had information they did not. The lesson is not that people are foolish; it is that a system which is right ninety-nine times trains you to stop checking on the hundredth.

KEY IDEA

Human oversight is a control only if the reviewer **can** actually disagree. That requires: enough time per case; access to the inputs, not just the output; a genuine signal of the model’s uncertainty; training on the system’s known failure modes; and **organisational cover** for overruling it. The EU AI Act writes human oversight into the high-risk regime as a design requirement (Chapter 14). Most deployments satisfy the letter and miss every one of those five conditions.

10.2 Distribution shift: models decay

A model is a claim about a distribution, and distributions move. Populations change, behaviour adapts, upstream systems get replaced, a pandemic arrives, a competitor exits the market and its customers appear in your data. Performance degrades quietly, and it **rarely degrades evenly**: the groups nearest the decision boundary and the groups with the least data feel it first and hardest.

There is also **adaptation**: once people know a model is being used, they change their behaviour to score well, which invalidates the relationship the model learned. This is Goodhart again, arriving from the outside. Credit-repair services, CV keyword optimisation and search-engine optimisation are all industries built on the gap between a proxy and a goal.

✓ **IN PRACTICE: Monitoring that is worth the name**

- Track performance **by subgroup over time**, not just in aggregate. Aggregate monitoring will miss a group-specific collapse until it is large enough to move the mean.
- Monitor the **inputs** as well as the outputs: shifts in feature distributions arrive before shifts in measured accuracy, which depend on labels you may only get months later.
- Set explicit **review triggers**, not just dashboards: a defined disparity threshold that mandates a human investigation, with a named owner.
- Log decisions and the model version that made them. When something goes wrong eighteen months later, this log is the difference between an investigation and a guess.

10.3 The loop: when a prediction makes its own future

The most consequential deployment problem is that outputs become inputs.

Take resource allocation from a risk model. The model predicts elevated risk in some area, on historical records. The organisation acts: patrols, inspections or audits are concentrated where the score is high. More is found where more was looked for, and the finding is **real** in each individual case; what is not real is the **comparison** with places nobody visited. Those findings are recorded and become next cycle's training data. The model is now measuring its own past decisions and presenting them as evidence about the world.

EVERYDAY ANALOGY

Point a microphone at the speaker it feeds and you get a howl within seconds. Neither component is faulty. The fault is the **path from output back to input**, and no improvement to the microphone will fix it. Deployment feedback loops are the same phenomenon at a slower frequency.

The same structure appears widely: recommender systems that shape the preferences they claim to predict; hiring pipelines whose "successful hire" data comes only from people the last model let through; lending models that never observe how refused applicants would have repaid (the selective-labels problem of Chapter 4).

KEY IDEA

A single-pass disparity produces a fixed injustice. A **looped** disparity produces a growing one. Breaking the loop requires deliberate design: randomised or held-out exploration, collecting outcome data independently of the model's own decisions, capping how much of any retraining set may come from model-directed activity, and periodically evaluating against an external benchmark the model cannot influence.

QUIZ WATCH · WEEK 6

Expect: define automation bias and give two of the five conditions that make human oversight a real control; explain distribution shift and why it hits some groups first; and describe a feedback loop in four steps and say what breaks it.

11 Case Files

COVERS	Four documented harms, each read for its pipeline stage and its multiplier
TIME	About 20 minutes
IN BRIEF	Know two of these well enough to narrate in three sentences. Each shows an ordinary model multiplied by scale, speed and the absence of appeal.

The cases in this chapter are not curiosities. They are the reason the EU AI Act exists in its current form, and they share a shape: the model was unremarkable, and the **system** around it converted an ordinary error rate into a catastrophe.

▀ CASE FILE: The Dutch childcare benefits scandal (toeslagenaffaire)

The Dutch tax administration used risk models to select families claiming childcare benefit for fraud investigation. Among the features contributing to a family's risk profile was **dual nationality**. Selected families faced demands to repay large sums, often immediately and often on the basis of minor administrative errors, with the burden of disproving fraud falling on them.

Roughly **26,000 families** were wrongly accused. Many were driven into serious debt; a substantial number of children were placed into care. The Dutch data protection authority found the processing of nationality data unlawful and fined the tax administration. In January 2021 the **government resigned** over the affair.

Stages: the data (nationality as a feature), the deployment (no meaningful appeal, reversed burden of proof), and the objective (detect fraud, with no counterweight for the cost of a false accusation). **Multiplier:** scale plus the absence of any route to contest.

▀ CASE FILE: Robodebt, Australia

Between 2015 and 2019 the Australian government operated an automated debt-recovery scheme that compared annual tax data against fortnightly welfare reporting, and **averaged** the annual income across the year to infer overpayments. Income earned irregularly, which is the norm for the low-paid casual work the system's subjects typically did, generated debts that did not exist.

The scheme reversed the burden of proof onto recipients, many of whom paid debts they did not owe. It was found unlawful, settled for around 1.8 billion Australian dollars, and a Royal Commission reported in 2023 in scathing terms about the scheme's design and the disregard of internal legal advice.

Stages: the objective and the method (an averaging assumption known to be wrong for this population), and deployment (automated demands at scale). **Multiplier:** speed and reversed burden of proof. Note that this system barely involved machine learning: **the harm was in the automation of a decision, not in the sophistication of the model.**

▀ CASE FILE: The 2020 exam grading algorithm, United Kingdom

With examinations cancelled during the pandemic, an algorithm adjusted teacher-assessed grades using each school's historical performance distribution. Around **39 per cent** of grades were lowered relative to teacher assessment. Because the method leaned on school-level history, it systematically disadvantaged high-achieving pupils in large, historically lower-attaining state schools, while small cohorts, more common in private schools, were left closer to teacher assessment.

The policy was withdrawn within four days after national protest.

Stages: the objective (preserve the national grade distribution, rather than assess this pupil), the model (school history as a feature about an individual), and deployment (no appeal at the point it mattered). **Multiplier:** every pupil at once, on one morning, with university places attached.

► CASE FILE: Apple Card, and why “no violation found” is still instructive

In late 2019, allegations circulated that the Apple Card, underwritten by Goldman Sachs, offered women markedly lower credit limits than men with comparable finances. New York’s Department of Financial Services investigated, analysing underwriting data for roughly 400,000 applicants, and concluded that it **did not find unlawful discrimination** under fair-lending law. The regulator nonetheless criticised customer service and a lack of transparency, and used the report to argue for modernising credit scoring and anti-discrimination rules.

This case is in the booklet precisely because the finding was negative. Two lessons follow. **First:** individual anecdotes, however vivid, cannot establish a disparity; that requires the population data, which only the firm or a regulator holds. **Second:** the reason the dispute could not be settled publicly for eighteen months is that **nobody outside could inspect the model**. Opacity did not protect the firm from the accusation; it simply moved the resolution to a regulator, months later, at great cost. Transparency is not only an ethical obligation. It is also, sometimes, the cheapest defence available.

KEY IDEA

The recurring anatomy: an **ordinary** model, multiplied by **scale** (everyone at once), **speed** (faster than any human check), and **no route of appeal**. None of those three multipliers is a modelling property. All three are architectural and organisational choices, which means all three are yours to set.

QUIZ WATCH · WEEK 6

Expect one scenario-based item drawn from this chapter’s shape. Be able to narrate two cases in three sentences each, name the pipeline stage that failed, and name the multiplier. The Robodebt observation that the system barely used machine learning is a favourite discriminator between strong and weak answers.

12 Explainability, and What It Cannot Do

COVERS	Global and local explanation, interpretable-by-design versus post hoc, SHAP and LIME, and the limits of all of it
TIME	About 22 minutes
IN BRIEF	Explanation is the tool for catching proxies, and it is routinely oversold. An explanation describes a model; it does not justify a decision, and it can be unfaithful, contradictory, or deliberately gamed.

Week 1 promised that connectionism's signature weakness was opacity, and that Week 2 would inherit the problem. Here it is. A deep model's behaviour is distributed across millions or billions of parameters, none of which corresponds to a reason. Yet the person refused a loan, the regulator reviewing the refusal, and the engineer debugging the pipeline all need something they can read.

12.1 Two axes, four quadrants

Global versus local. A **global** explanation describes how the model behaves in general: which features matter overall, how the prediction varies with a feature across the population. A **local** explanation addresses a single case: why **this** output for **this** person. Local explanation is what an affected individual and a regulator both actually want, and it is the harder of the two.

Interpretable by design versus post hoc. An **interpretable** model is one whose mechanism a person can follow directly: a short decision list, a scoring card, a sparse linear model, a shallow tree. The model **is** the explanation. A **post-hoc** method builds a second, simpler artefact that describes the first: it approximates a black box, from outside, after the fact.

12.2 The two post-hoc methods you will use

LIME (Ribeiro and colleagues, 2016) explains one prediction by perturbing the input many times, observing how the model's output changes, and fitting a simple interpretable model, typically a sparse linear one, to those perturbations **in the neighbourhood of that point**. The output is a short list of the features that locally push the prediction one way or the other. It is model-agnostic, intuitive, and depends heavily on how the neighbourhood is defined.

SHAP (Lundberg and Lee, 2017) draws on cooperative game theory. Treat the features as players in a game whose payoff is the prediction, and ask how the payoff should be divided among them. **Shapley values** provide the unique allocation satisfying a set of desirable axioms, and SHAP adapts them to machine learning, with efficient exact algorithms for tree ensembles and approximations elsewhere. Its attraction is that the contributions **sum** to the difference between this prediction and the population base value, giving a complete, additive account.

EVERYDAY ANALOGY

A local explanation is a **receipt**, not a **recipe**. It is an itemised account of what was charged, produced after the transaction, by the shop. It tells you which items drove the total, which is genuinely useful. It does not tell you whether the shop should have sold you those items, and it was not produced by an independent party.

12.3 What explanation is genuinely good for

Do not read this chapter's caveats as a case against using these tools. Local explanation is **the single best practical technique for catching a proxy**. When a refusal is dominated by **postcode**, or a hiring score is dominated by a

variable that correlates with parenthood, an attribution method will usually show it, in seconds, on a laptop. That is how a great many real fairness problems are actually found. It also supports debugging, model comparison, stakeholder communication, and the “meaningful information about the logic involved” the GDPR requires (Chapter 14).

12.4 Four limits you must know

1. **An explanation describes the model, not the decision’s justification.** SHAP reports that postcode drove the refusal. It has no opinion about whether postcode **should** drive a refusal. The moral work is entirely yours, and reporting the attribution can feel like discharging it.
2. **Post-hoc explanations can be unfaithful.** They are approximations. A simple surrogate fitted near a point may be a poor description of a highly non-linear model even slightly further away, and it will look equally confident either way.
3. **Methods disagree.** Run LIME and SHAP, or two SHAP variants with different background distributions, on the same case and you can get different feature orderings. The literature calls this the **disagreement problem**, and it has no principled tie-breaker. If your process is “produce an explanation”, disagreement is invisible; if your process is “produce two and compare”, it is informative.
4. **They can be gamed.** Published attacks construct models that behave one way on real inputs and another on the perturbed inputs an explainer generates, yielding innocuous explanations for discriminatory behaviour. Explanation is therefore not an **audit**: an adversary who controls the model can, in principle, control the explanation.

KEY IDEA

A plausible explanation increases trust whether or not it is faithful, which makes explainability a candidate for the ELIZA effect at institutional scale. Week 1’s standing rule applies unchanged: **test on substance, never on fluency**. A confident, readable explanation is fluent output. Its faithfulness is a separate, checkable property, and you should check it.

DEEPER DETAIL: Rudin’s argument, and the honest response

In 2019 Cynthia Rudin published an argument that has only grown in influence: for high-stakes decisions about people, **stop explaining black boxes and use interpretable models instead**. Her case has three parts. Post-hoc explanations are approximations and can mislead, so a system justified by one may be justified on a fiction. The assumed accuracy cost of interpretable models is frequently **assumed rather than measured**, and on much structured, tabular data a carefully built interpretable model performs comparably. And the existence of a plausible explanation provides **cover** for deploying systems that should not be deployed.

The honest counter-argument is that for images, audio and natural language the performance gap is real and large, and post-hoc tools are what exists. Both can be true, and they point at the same professional practice: **measure the gap before you accept it**. Fit the interpretable baseline. If it costs two points of accuracy on a decision that changes someone’s life, that is a trade you must argue explicitly, with the numbers, rather than assume. In many projects the baseline is competitive, and discovering that is worth a week of your time.

✓ IN PRACTICE: Explanation, done properly

- Fit an **interpretable baseline** early, and record the performance gap. It is your evidence either way.
- Generate explanations with **at least two methods** and compare. Agreement is reassurance; disagreement is a finding.

- Check **faithfulness**, not just plausibility: does removing or perturbing the features the explanation named actually change the prediction as predicted?
- Write the **user-facing** version. If you cannot state, in a paragraph a refused applicant could read, which factors moved their score and in which direction, you do not yet have a deployable system.

QUIZ WATCH · WEEK 6

Reliable: distinguish global from local and interpretable from post hoc, with an example of each; explain LIME and SHAP in one sentence each, including SHAP's game-theoretic basis; and give three limits of post-hoc explanation. A strong answer connects the fourth limit to the ELIZA effect from Week 1.

13 Accountability: Who Answers

COVERS	The many-hands problem, contestability, and why accountability is a property of organisations rather than models
TIME	About 12 minutes
IN BRIEF	“The algorithm decided” is a transfer of responsibility to something that cannot hold any. This chapter is about what fills the gap: records, named owners, and a route of appeal.

13.1 The modern “computer says no”

When an automated system harms someone, the sentence that follows is remarkably consistent across countries, sectors and decades: **the system flagged it, the algorithm decided, that is what the model said**. It sounds like an explanation. It is a transfer of responsibility to an entity that cannot bear any, and which has no assets, no duty of care and no capacity to be sorry.

The reason it is so tempting is the **many-hands problem**. Consider who contributed to a single automated refusal: the team that collected the data; the contractor who labelled it; the product manager who chose the target variable; the engineer who trained the model; the risk function that set the threshold; the procurement team that bought a vendor component; the caseworker who accepted the output under a throughput target; and the executive who approved the deployment. Every one contributed. No one decided. Diffusion of this kind is not usually a conspiracy; it is the default state of any organisation that has not deliberately arranged otherwise.

KEY IDEA

Accountability is not a property of a model. It is a property of **organisations and their records**. The concrete test is a question: for each of the seven pipeline stages, **can you name the person who made the decision and find the document in which they made it?** Where the answer is no, you have not found a philosophical puzzle; you have found a missing document, and missing documents can be created.

13.2 Contestability: the property that most systems lack

Look again at the case files of Chapter 11 and notice what unites them. In each, the decisive failure was not the error rate. It was that the affected person **could not effectively contest the outcome**: the burden of proof was reversed, the appeal arrived too late to matter, or nobody could explain what had happened.

Contestability means an affected person can find out that an automated system was involved, obtain an intelligible account of why, present information the system did not have, and reach a human with the authority to change the outcome, within a timescale in which changing it still helps. It is an architectural requirement, not a customer-service policy, and it is the single most effective mitigation in this booklet, because it works even when your fairness analysis was wrong.

✓ IN PRACTICE: The accountability minimum

- A **named owner** for the system, and named decision-makers for the target variable, the fairness criterion and the threshold.
- A **decision log**: what was decided, by which model version, on what inputs, and when. This is what makes an investigation possible eighteen months later.
- A **documented appeal route**, with a stated timescale, that reaches a human with authority to reverse the outcome.

- An **incident process**: how a suspected harm gets reported, triaged, escalated and recorded, including by people outside the organisation.
- A **retirement plan**. Systems that cannot be turned off are systems that will keep causing whatever they cause.

QUIZ WATCH · WEEK 6

Likely: explain the many-hands problem in two sentences; define contestability and list three of its components; and state why accountability is described as a property of organisations rather than of models.

14 Regulation as an Engineering Requirement

COVERS	The EU AI Act, the GDPR, the case law that matters, and how AI is actually regulated in Ireland
TIME	About 25 minutes
IN BRIEF	Read as a specification rather than an obstacle, the law asks for things you would want anyway: know your data, evaluate by subgroup, log decisions, keep a human who can overrule, and be able to explain.

A caution before anything else. This chapter is a **technical** orientation for engineers, not legal advice, and it describes a moving target: the EU timetable has already been amended once, in 2026, and may be amended again. Verify dates against a primary source before relying on them in an assignment or a project.

14.1 The AI Act's shape: risk tiers, and use rather than technique

The EU Artificial Intelligence Act, Regulation (EU) 2024/1689, entered into force on 1 August 2024. Its logic is **product safety**: obligations scale with the risk of the **use**, not the sophistication of the technique. The same model can be unregulated in a game and high risk in a hiring tool.

- **Unacceptable risk: prohibited.** Social scoring by public authorities, certain manipulative or exploitative systems, untargeted scraping of facial images to build recognition databases, and further categories added by later amendment.
- **High risk: regulated.** The tier that will govern most of the careers this programme leads to: employment and worker management, creditworthiness, education, access to essential public and private services, law enforcement, and safety components of regulated products. Obligations include risk management, data governance, technical documentation, logging, human oversight, accuracy and robustness requirements, and conformity assessment before market.
- **Transparency obligations.** Applying across tiers: people must be told when they are interacting with an AI system, and synthetic media must be marked as such.
- **Minimal risk.** Everything else, which is most things, and which carries no specific obligations.

§ THE LAW SAYS: The compliance timetable, as amended in 2026

The Act phases in by obligation rather than all at once. As matters stand after the **Digital Omnibus on AI**, which received final Council approval on 29 June 2026:

- **1 August 2024**, the Act enters into force.
- **2 February 2025**, prohibitions apply; the **AI literacy** duty on providers and deployers begins. (This lecture is, formally, part of somebody's compliance obligation.)
- **2 August 2025** (obligations for general-purpose AI models, governance structures and penalties.
- **2 August 2026**) transparency obligations apply.
- **2 December 2026** (newly added prohibitions take effect, and marking duties extend to systems already on the market.
- **2 December 2027**) the high-risk regime for stand-alone (Annex III) systems. **This was moved from 2 August 2026** by the Omnibus, after implementation, standards and national authority designations fell behind.
- **2 August 2028**, high-risk obligations for AI embedded in regulated products.

Penalties at the top of the range reach tens of millions of euro or a percentage of worldwide annual turnover, with the highest tier reserved for prohibited practices.

14.2 The paradox the law has just had to resolve

There is a genuine conflict at the heart of fairness engineering, and it is worth understanding because it explains an otherwise puzzling amendment.

To demonstrate that a model disadvantages women, you need the applicants' sex. To test for racial disparity, you need race. To test for disability discrimination, you need disability status. **Every fairness metric in Chapter 8 requires the protected attribute.** Yet Article 9 of the GDPR prohibits, by default, the processing of exactly these special categories of personal data. The result, for years, was that fairness testing was often blocked by the rules designed to protect the same people, and “we are not permitted to collect that” became a widely used and partly sincere reason for not measuring disparity at all.

§ THE LAW SAYS: Special-category data for bias detection

The AI Act already contained a narrow exception, in Article 10(5), permitting processing of special-category data where **strictly necessary** for bias monitoring, detection and correction **in high-risk systems**. The 2026 Omnibus widened this considerably, providing a legal basis for providers and deployers of AI systems more generally to process such data for bias detection and correction, subject to a strict-necessity test and safeguards: preferring non-sensitive or synthetic alternatives where they exist, pseudonymisation, access controls, no onward sharing, and deletion when no longer needed.

Two points of interpretation matter for you. It creates a **permission**, not an obligation: nothing here requires you to perform bias detection. And it is narrow: the data may be used **for that purpose**, not repurposed for modelling.

KEY IDEA

Read that amendment as an engineering instruction: **the legal excuse for not measuring disparity has narrowed sharply**. Plan the audit data, the safeguards and the deletion schedule at design time. “We could not lawfully measure it” is a rapidly ageing defence.

14.3 The GDPR provisions you will actually meet

Article 22 restricts decisions based **solely** on automated processing that produce legal or similarly significant effects, with exceptions and, where they apply, safeguards including the right to obtain human intervention, to express a point of view, and to contest the decision.

Article 15(1)(h) gives the data subject a right to **meaningful information about the logic involved** in such processing, along with its significance and envisaged consequences.

Two judgments of the Court of Justice have sharpened both.

§ THE LAW SAYS: SCHUFA (C-634/21, December 2023) and Dun & Bradstreet Austria (C-203/22, February 2025)

In **SCHUFA**, the Court held that producing a credit score can itself constitute an automated decision under Article 22, where the score effectively determines whether a third party will contract with the person. The practical consequence: **“we only supply a score, our client makes the decision”** is not a reliable defence for a scoring provider.

In **Dun & Bradstreet Austria**, the Court addressed what “meaningful information about the logic involved” requires. It is the **procedure and principles actually applied** to the person's data to reach the result, communicated concisely, transparently and intelligibly. Notably, it is satisfied **neither** by disclosing a complex mathematical formula or the algorithm itself, **nor** by a mere description of every step. And trade secrecy does not extinguish the right: where a controller claims it, the material goes to the competent court or supervisory authority to weigh against the data subject's rights.

KEY IDEA

The design instruction is straightforward: **build for the explanation you will owe**. If you cannot state, in a paragraph a refused applicant could read, which factors moved their score and in which direction, you do not have a deployable system. Notice that this is the same requirement Chapter 12 arrived at on purely engineering grounds.

14.4 How this lands in Ireland

Ireland chose a **distributed** model of oversight rather than a single AI regulator. Around fifteen existing sectoral authorities were designated as competent authorities and market surveillance authorities, each supervising AI within its own domain, with a central **National AI Office** providing coordination, acting as single point of contact, and running a national regulatory sandbox. Domestic implementing legislation has been progressing through a General Scheme of a Regulation of Artificial Intelligence Bill.

The consequence for a graduate of this programme is that **your regulator depends on your sector**: the Data Protection Commission where personal data is processed, the Central Bank of Ireland in financial services, Coimisiún na Meán for media and online platforms, the Workplace Relations Commission for employment, and so on. There is an argument for this approach (sectoral expertise, existing enforcement powers) and an argument against it (coordination cost, and the risk of inconsistent interpretation), and it is a legitimate topic for your final project's critical brief.

DEEPER DETAIL: Other frameworks you will meet in industry

Beyond the binding EU regime, three voluntary frameworks recur in job descriptions and procurement documents. The **NIST AI Risk Management Framework** (United States, 2023) organises practice around four functions: govern, map, measure, manage; it is non-binding, widely referenced, and a reasonable skeleton for a project's risk section. **ISO/IEC 42001** (2023) specifies an AI management system, in the same family as ISO 27001 for information security, and is certifiable, which is why procurement teams like it. The **OECD AI Principles** and the UNESCO recommendation supply the value vocabulary that the EU text and most corporate policies draw on.

For your purposes, the useful observation is that these frameworks converge on the same operational core: know your data, define intended use, assess risk before deployment, evaluate by subgroup, keep humans able to intervene, log everything, and monitor after release. If you do those seven things, most of the paperwork for most of the frameworks is a formatting exercise.

QUIZ WATCH · WEEK 6

Expect: name the four AI Act risk tiers with one example each; state one obligation on high-risk systems; explain what Article 22 restricts and what Article 15(1)(h) grants; explain the fairness measurement paradox and how the law changed in response; and name two Irish competent authorities and their domains. Do not memorise every date, but know that the high-risk deadline **moved**, and why.

15 The Practice

COVERS	Documentation, tooling, and the three families of mitigation, with the cost of each
TIME	About 18 minutes
IN BRIEF	What you actually do on Monday: datasheets, model cards, disaggregated evaluation, and pre-, in- and post-processing mitigations, each with a stated price.

15.1 Two documents that are not bureaucracy

Datasheets for datasets (Gebru and colleagues) accompany a dataset with the answers to a standard set of questions: why it was created, who funded it, what the instances represent, how it was collected, from whom, with what consent, how it was cleaned, whether it contains sensitive content, what tasks it should be used for and, crucially, what tasks it should **not**. It is written by whoever collected the data, while they still remember.

Model cards (Mitchell and colleagues) accompany a model with: intended use and out-of-scope use, the training data, evaluation data, **performance disaggregated by group**, known limitations, ethical considerations, and, for this module's purposes, the fairness criterion chosen and the reasoning behind it. One page, and it is the artefact that a regulator, a buyer, an auditor and your successor all ask for first.

KEY IDEA

Neither document is paperwork. Both are **the mechanism by which a decision acquires a name**. An undocumented system cannot be audited, cannot be safely inherited, and cannot be defended when challenged. This is also the cheapest thing in this chapter: a model card takes an afternoon.

15.2 Three places to intervene, and what each costs

Stage	What it does	What it costs
Pre-processing	Reweight, resample, or repair the training data: massage the distribution before the model sees it.	Model-agnostic and treats causes rather than symptoms. But it alters data you may be required to retain unmodified, and it can be hard to explain to an auditor.
In-processing	Add fairness constraints or penalties to the objective; adversarial debiasing; worst-group (distributionally robust) objectives.	The most principled place to act, and the trade is explicit in the loss. Requires control of training, and can be less stable.
Post-processing	Adjust outputs or thresholds after the fact, potentially per group.	Works on models you cannot retrain, including vendor components. Frequently the legally riskiest option in the EU, since per-group thresholds use the protected attribute at decision time.

Notice the geometry against Chapter 3's pipeline: post-processing patches **stage six** for a problem usually created at **stage three or four**. It is the most common intervention because it is the latest and cheapest to implement, and it is the least effective for the same reason. **The cheapest real fix is always the earliest one.**

15.3 The metrics, written out

Every criterion in Chapter 8 is arithmetic on a confusion matrix. The Week 2 lab implements them in numpy rather than importing a fairness library, because once you have seen how short they are, “the tool says the model is fair” stops being an acceptable answer:

```
def group_metrics(y, s, t, label):
    yhat = (s >= t).astype(int)          # the threshold IS the decision
    tp, fp, fn, tn = confusion(y, yhat)
    return GroupMetrics(
        selection_rate = _safe_div(tp + fp, len(y)),    # demographic parity
        tpr             = _safe_div(tp, tp + fn),      # equal opportunity
        fpr             = _safe_div(fp, fp + tn),      # equalised odds
        ppv             = _safe_div(tp, tp + fp),      # predictive parity
    )
```

Note `_safe_div`, which returns nan rather than zero when a denominator is empty. Returning zero would be a lie that flatters the model: a group with no positive cases would appear to have a perfect false-negative rate.

15.4 What the lab actually finds

Running `mitigate.py` on the default seed compares four remedies against doing nothing. The table is the argument of this chapter in one screen:

strategy	accuracy	sel_gap	tpr_gap	fpr_gap	ppv_gap
baseline (nothing done)	0.6986	0.2025	0.2053	0.1221	0.0792
drop postcode_score	0.6839	0.2731	0.2728	0.2024	0.0459
reweighting (pre-processing)	0.6818	0.1028	0.1146	0.0201	0.1260
per-group thresholds (post, selection)	0.6836	0.0003	0.0297	0.0982	0.1629
per-group thresholds (post, FPR)	0.6893	0.0849	0.0944	0.0005	0.1296
fix the label (stage 3)	0.6979	0.0326	0.0333	0.0316	0.0087

Four things to take from it. **Dropping the obvious proxy makes the disparity worse**, because the model leans harder on the proxies that remain; there is no single column to delete. **Reweighting** helps and costs a little accuracy, and quietly worsens the predictive-parity gap. **Per-group thresholds** zero whichever gap they were tuned for and inflate the others, which is Chapter 9 in production form. And **fixing the label** collapses nearly every gap at an accuracy cost of 0.0007, because stage three is where the harm was introduced.

15.5 The tooling

All of the following are free and mature enough for coursework and for production: **Fairlearn** (metrics and mitigation algorithms, well integrated with scikit-learn), **AI Fairness 360** (a broad library of metrics and mitigations with a research lineage), the **What-If Tool** (interactive counterfactual and threshold exploration), and **SHAP** and **LIME** for explanation. The Week 2 lab uses this stack, and the Bias & Fairness Explorer is a simplified version of what these tools do.

✓ IN PRACTICE: A realistic responsible-AI workflow

- 1. Before data.** Write the goal in plain language. Identify affected groups and plausible harms. Decide, provisionally, on a fairness criterion and record why.
- 2. At data.** Produce or obtain a datasheet. Build the population table. Establish lawful basis and safeguards for any protected attributes needed for testing.
- 3. At label and target.** Write the two-sentence goal-versus-proxy comparison. List divergences and who they fall on.
- 4. At training.** Fit the interpretable baseline. Fit the intended model. Report both aggregate **and** worst-group performance.

5. **At threshold.** Sweep it. Produce the trade-off curve across your candidate criteria. Choose in a meeting, with a written record and named participants.
6. **Before release.** Model card. Explanation samples. Human-oversight design, with the five conditions of Chapter 10. Appeal route with a stated timescale.
7. **After release.** Subgroup monitoring, input drift monitoring, review triggers with named owners, decision logs, and an incident process.

QUIZ WATCH · WEEK 6

Likely: state what a model card contains and why disaggregated performance is on it; name the three families of mitigation with one example and one cost each; and explain in one sentence why post-processing is popular and yet the weakest option.

16 A Field Guide: Seven Questions for Any Model You Build

COVERS	The whole week, operationalised as a checklist for anything you ship
TIME	About 12 minutes, then use it forever
IN BRIEF	Week 1 gave seven questions for anything you read. These are the seven for anything you build, each anchored to a chapter you now know.

History you cannot use is trivia, and ethics you cannot apply is worse: it is decoration. This chapter converts the booklet into an instrument. Bring it to the Week 2 lab, to every project in the remaining ten weeks, and to the critical brief of your final project, where it is close to a mark scheme.

- 1. Who is in this data, and who is missing?** The representation question (Chapter 4). Produce the population table and compare it against the population the system will serve. Test at intersections, where the largest disparities hide and the counts are smallest. If a group is absent, say so in the model card rather than discovering it in production.
- 2. Who wrote the label, and what were they really recording?** The label question (Chapter 5). Complete the sentence “our label actually records...”. If the honest completion differs from the column’s name, that gap is your principal risk and it belongs in writing.
- 3. What is the target a proxy for, and where does the proxy break?** The objective question (Chapter 6), and the one with the largest expected payoff. State the goal and the target as two separate sentences. List the mechanisms by which they diverge, and ask, for each, whether it falls unevenly on a group. This is the question that would have caught the Obermeyer case before a line of code was written.
- 4. Who bears a false positive, and who bears a false negative?** The allocation question (Chapter 7). Two sentences, in plain language, about real people. If the two errors fall on different groups with different severity, you are allocating harm and you need a stated reason.
- 5. Which fairness criterion am I committing to, and what am I giving up?** The impossibility question (Chapters 8 and 9). You cannot satisfy all of them; pretending otherwise is the only unambiguously wrong answer. Name the criterion, name what it costs, name who agreed it.
- 6. Can the affected person get an explanation and contest the outcome?** The contestability question (Chapters 12 and 13, and the law in 14). Can they discover that a system was involved, obtain an intelligible account, add information, and reach a human with authority to change it, in time for the change to matter?
- 7. What happens when this output becomes next year’s training data?** The loop question (Chapter 10). Trace the path from your decisions to your future datasets. If it closes, say how you will break it: held-out exploration, independent outcome collection, caps on model-directed data, or external benchmarks the model cannot influence.

TRY IT YOURSELF

Standing exercise for the semester, and reliable forum gold: take one **deployed** system you personally use each week, and run it through all seven questions in under two hundred words. By Week 6 this is quiz preparation. By Week 12 it is a draft section of your final project’s critical brief, where the responsibility analysis is explicitly marked.

17 The Revision Core: The Whole Week on Two Pages

If you memorise nothing else for Week 6, memorise this chapter.

17.1 The three senses of bias

Statistical (an estimator systematically off; morally neutral). **Inductive** (the assumptions that make learning possible; a virtue, and the subject of Weeks 4 and 5). **Societal** (systematic disadvantage to a group; the only one that can wrong somebody). Never write “bias” without the qualifier.

17.2 The seven stages, and what leaks at each

Stage	What it is	What leaks, and the exhibit
1. World	The historical record you inherit	Historical bias. The data is accurate; the world was not fair. Arrests, not offences.
2. Data	Who was sampled, and how measured	Representation and measurement bias. Gender Shades : 0.8% error to 34.7% across subgroups, hidden by a skewed benchmark.
3. Labels	Who decided what “correct” means	Label and annotator bias. Amazon : the label was “who we hired before”; deleting words did not delete proxies.
4. Objective	The measurable stand-in for the goal	Proxy divergence and Goodhart. Obermeyer : cost as a proxy for need; 17.7% to 46.5% on fixing the target.
5. Model	What the loss rewards	Aggregation: the majority writes the loss. Worst-group performance diverges from the average. Compression makes it worse (Week 9).
6. Threshold	One number turning a score into a decision	Error allocation. One identical rule, two selection rates. Not choosing is choosing 0.5.
7. Deployment	People, incentives and time pressure	Automation bias, drift, and no appeal route. Then the arrow back: outputs become tomorrow’s inputs.

17.3 The four fairness families

Demographic parity (equal selection rates; equality of outcome; ignores real base-rate differences). **Equalised odds and equal opportunity** (equal error rates; best where the decision is punitive; trusts the label). **Calibration and predictive parity** (a score means the same for everyone; what makes a score usable; forces unequal errors). **Individual and counterfactual** (similar people similarly treated; needs a similarity metric or a causal model you probably lack).

17.4 The impossibility result

If base rates differ between groups and the classifier is imperfect, **calibration**, **equal false-positive rates** and **equal false-negative rates** cannot all hold. Any two can. The escapes are equal base rates or a perfect classifier,

and you will have neither. COMPAS: ProPublica measured error rates, the vendor measured calibration, both were correct. Unlike XOR, **there is no escape route**: the theorem relocates the work from “is it fair?” to “which fairness, on what account of the harm, at what measured cost, agreed by whom?”

17.5 Eight one-line lessons

1. A model that allocates something is not a measurement but an intervention; a faithful model of an unfair world is an accurate model, and deploying it is a decision to continue.
2. Removing a protected attribute removes your ability to measure the disparity, not the disparity (fairness through unawareness; Amazon).
3. The most consequential fairness decision is usually the choice of target variable, made months before training, by people not thinking about ethics (Obermeyer).
4. An aggregate accuracy figure is an average over people, and averages hide order-of-magnitude differences (Gender Shades).
5. Not choosing a threshold is choosing the library default, and the threshold allocates every error in the system.
6. Several reasonable fairness criteria are provably incompatible; the professional act is choosing one and documenting why.
7. An explanation describes the model; it does not justify the decision, and it can be unfaithful, contradictory or gamed.
8. The case files share an anatomy: an ordinary model multiplied by scale, speed, and the absence of appeal. All three multipliers are your design choices.

QUIZ WATCH · WEEK 6

The Week 6 quiz draws its Week 2 items from exactly this chapter’s territory: the three senses, the seven stages as a diagnostic scaffold, the four families with their trade-offs, the impossibility result stated precisely, and two case files narrated accurately. If you can reconstruct this chapter from memory onto two blank pages, Week 2 revision is complete.

18 Glossary in Plain English

Terms are ordered by first appearance rather than alphabetically, so the glossary doubles as a skeleton retelling. Each entry gives the plain meaning first, then the chapter where it lives.

Term	Plain-English meaning	Ch.
Statistical bias	An estimator systematically off in one direction. The bias in the bias-variance trade-off. Technical, not moral.	2
Inductive bias	The assumptions a model brings before seeing data. Without some, learning is impossible. Weeks 4 and 5 add it deliberately.	2
Societal bias	Systematic disadvantage to a group of people, intended or not. The only sense in which a model can wrong somebody.	2
Allocative harm	A resource or opportunity withheld: the loan, the interview, the parole. Countable, and what nearly all metrics measure.	2
Representational harm	A system encoding who counts as normal, with no single denial to point at. Cumulative, and largely unmeasured.	2
Historical bias	The data is accurate and the world it recorded was unjust. No collection process fixes it.	4
Representation bias	Some groups are under-sampled, so the loss barely notices them and the average conceals it.	4
Measurement bias	The instrument itself behaves differently across groups, so the same number does not mean the same thing.	4
Disaggregated evaluation	Reporting performance separately for each group and intersection. The cheapest responsible-AI practice there is.	4
Selective labels	Outcomes exist only for cases a previous decision selected, so the model partly measures agreement with the old policy.	4
Label bias	The “ground truth” column records a human judgement, and inherits its definition, pressures and point of view.	5
Proxy variable	A feature that carries information about a protected attribute without being it: postcode, first name, employment gaps, device type.	5
Fairness through unawareness	Dropping the protected attribute and calling the result neutral. Fails because proxies remain, and destroys your ability to measure.	5
Target variable	The column you actually predict. Usually a measurable stand-in for the thing you care about.	6
Proxy target	The measurable stand-in itself: cost for need, re-arrest for reoffence, retention for job performance.	6
Goodhart’s law	When a measure becomes a target it stops being a good measure. Machine learning amplifies the effect by optimising hard.	6

Aggregation bias	One model fitted across groups whose feature-outcome relationships differ; the majority writes the loss.	7
Worst-group performance	Accuracy on the worst-served group rather than the mean. Report it alongside the average, always.	7
Threshold	The number converting a score into a decision. Not learned, chosen, and it allocates every error in the system.	7
False positive / false negative	Wrongly flagged / wrongly missed. Different harms, borne by different people, traded against each other by the threshold.	7
Demographic parity	Equal selection rates across groups. Equality of outcome; ignores real differences in the outcome being predicted.	8
Equalised odds	Equal true-positive and false-positive rates. Equal opportunity is the weaker version: equal true-positive rates only.	8
Calibration	A score of 0.7 means a 0.7 chance for everybody. What makes a score usable at all; collides with equal error rates.	8
Predictive parity	Equal positive predictive value across groups: the thresholded version of calibration.	8
Individual fairness	Similar individuals treated similarly. Moves the difficulty into defining “similar”, which is the moral question in disguise.	8
Counterfactual fairness	The decision would be the same had the person belonged to another group. Conceptually sharpest; needs a causal model.	8
Impossibility result	With unequal base rates and an imperfect classifier, calibration and equal false-positive and false-negative rates cannot all hold.	9
Base rate	How often the outcome actually occurs in a group. Differences in it are what force the trade-offs, and may themselves be label artefacts.	9
Automation bias	People over-trust machine output, especially when rushed or measured on throughput. Why “a human reviews it” is usually not a control.	10
Distribution shift	The world moves away from the training distribution, unevenly, hurting the least-represented groups first.	10
Feedback loop	Outputs become inputs: you find what you looked for, log it, retrain on it. Small disparities compound instead of washing out.	10
Contestability	The affected person can learn a system was used, get an intelligible account, add information, and reach a human who can change it in time.	13
Global vs local explanation	How the model behaves in general, versus why this output for this person. Regulators and applicants want local.	12
Interpretable by design	The model itself is readable: short decision lists, scorecards, sparse linear models. No second artefact needed.	12

LIME	Perturb around one point, fit a simple local surrogate, report which features pushed the prediction. Model-agnostic; neighbourhood-sensitive.	12
SHAP	Shapley values from cooperative game theory, splitting a prediction among features so contributions sum to the gap from the base value.	12
Disagreement problem	Different explanation methods, or the same one differently configured, give different feature orderings on the same case.	12
Many-hands problem	Everyone contributed to an automated decision and nobody decided it. Why accountability needs named owners and documents.	13
Datasheet for a dataset	A standard record of how data was collected, from whom, with what consent, and what it must not be used for.	15
Model card	A one-page record of intended use, training data, disaggregated performance, limitations, and the fairness criterion chosen.	15
Pre-, in-, post-processing	Fix the data, fix the objective, or fix the output. Earliest is cheapest and most effective; latest is most common and most legally exposed.	15
EU AI Act	Risk-tiered EU regulation of AI by use : prohibited, high risk, transparency, minimal. In force 2024, phasing in to 2028.	14
GDPR Article 22	Restricts decisions based solely on automated processing with legal or similarly significant effects, with safeguards including human intervention.	14
Meaningful information about the logic	Article 15(1)(h): the procedure and principles actually applied, concise and intelligible. Neither a formula dump nor a hand-wave.	14

19 Question Bank: Practising for Week 6

The Week 6 quiz mixes auto-graded items with short structured responses and a brief applied component. Everything below is written in those registers. Answers are deliberately not printed; every question is answerable from a named chapter, and hunting the answer is the revision.

19.1 Short answer: two to three sentences each

1. Define the three senses of the word “bias” and say which one Week 5 deliberately increases. (Chapter 2)
2. Explain why a model with no inductive bias cannot learn. (Chapter 2)
3. Distinguish allocative from representational harm, with one example of each, and state which kind standard fairness metrics measure. (Chapter 2)
4. List the seven pipeline stages in order. (Chapter 3)
5. Explain in one sentence why the feedback arrow makes disparities compound rather than dilute. (Chapters 3, 10)
6. Distinguish historical, representation and measurement bias. (Chapter 4)
7. State the Gender Shades finding, including the best and worst subgroup error rates, and explain why existing benchmarks had not detected it. (Chapter 4)
8. Give two reasons why “just collect more data” can fail as a remedy for representation bias. (Chapter 4)
9. Define the selective labels problem and give one domain in which it arises. (Chapter 4)
10. Explain what the label “reoffended” actually records, and why that matters for a risk model. (Chapter 5)
11. Narrate the Amazon recruiting case in three sentences and state the correct lesson from it. (Chapter 5)
12. Explain why fairness through unawareness fails, naming at least two proxy variables. (Chapter 5)
13. Define a proxy target and explain why proxies are unavoidable rather than merely careless. (Chapter 6)
14. State Goodhart’s law and explain why machine learning amplifies it. (Chapter 6)
15. Narrate the Obermeyer case in three sentences, including both percentages, and state which pipeline stage failed. (Chapter 6)
16. Explain aggregation bias and name two mitigations with their costs. (Chapter 7)
17. Explain how one identical threshold applied to two groups can produce very different selection rates. (Chapter 7)
18. Explain the phrase “not choosing a threshold is choosing 0.5”. (Chapter 7)
19. For a cancer screening system, say which error you would prefer to make and why; then do the same for a benefit-fraud flag. (Chapter 7)
20. Define demographic parity and give one context where it is defensible and one where it is not. (Chapter 8)
21. Distinguish equalised odds from equal opportunity. (Chapter 8)
22. Explain what calibration guarantees and why a decision-maker cares about it. (Chapter 8)
23. Explain why individual fairness moves the difficulty rather than removing it. (Chapter 8)
24. State the impossibility result precisely, including the two conditions under which the trade-off disappears. (Chapter 9)
25. Explain in plain language why calibration forces unequal error rates when base rates differ. (Chapter 9)
26. Contrast the impossibility result with the XOR critique of 1969 on the specific question of whether an escape route exists. (Chapters 9; Week 1 Chapter 6)
27. Define automation bias and give three conditions that must hold for human oversight to be a genuine control. (Chapter 10)
28. Describe a feedback loop in four steps and name one technique that breaks it. (Chapter 10)
29. Explain the many-hands problem and the concrete test for whether accountability exists. (Chapter 13)
30. Distinguish global from local explanation, and interpretable-by-design from post hoc. (Chapter 12)
31. Explain SHAP in one sentence, naming its game-theoretic basis, and give one limitation. (Chapter 12)

32. Summarise Rudin’s argument and give the strongest counter-argument to it. (Chapter 12)
33. Name the four risk tiers of the EU AI Act with one example each. (Chapter 14)
34. Explain the fairness measurement paradox and how EU law changed in 2026 in response. (Chapter 14)
35. State what a model card contains and why disaggregated performance belongs on it. (Chapter 15)
36. Name the three families of mitigation, with one example and one cost each, and say why the latest is the weakest. (Chapter 15)

19.2 Diagnostic: name the stage, justify in two sentences

For each scenario (all invented), identify the pipeline stage that most likely failed and the documented case it echoes.

1. A speech-recognition product reports 95% accuracy and is unusable for a large minority of its users in one region, none of whom appear in the evaluation set.
2. A model predicting “likelihood of needing intensive support” is trained on how much support each person received last year.
3. A university deploys a plagiarism-risk model trained on cases that were previously investigated and upheld.
4. A hiring model is retrained annually on the CVs of everyone hired since it was deployed.
5. A lender drops gender from its features, reports that the model is therefore non-discriminatory, and declines to compute approval rates by gender.
6. A fraud model’s default cut-off is left at 0.5 because “that is the standard”, and the flagged population turns out to be concentrated in two postcodes.
7. A vendor states that its risk instrument cannot be racially biased because a given score corresponds to the same reoffence rate in every group.
8. A caseworker with a ninety-second-per-case target agrees with the model’s recommendation 98% of the time, and this is reported as evidence of model accuracy.

19.3 Concept application: one paragraph each

1. A product manager proposes predicting “customer lifetime value” to decide who receives priority support. Write the goal-versus-proxy analysis, and name the group most likely to be disadvantaged and why.
2. You are told to “make the model fair” for a loan product. Write the reply you would send, naming the criteria available, the trade-off you cannot avoid, and the three things you need from the business before you can proceed.
3. Take any AI system you used this week and run it through four of the seven questions in Chapter 16.
4. Your team wants to ship a quantized on-device version of a model to improve privacy. Using Chapters 7 and 15, state the fairness risk this introduces and the specific evaluation you would demand before release.
5. A colleague presents a SHAP plot as evidence that a model is not discriminating. Explain, in one paragraph, what the plot does and does not establish.

19.4 Discussion prompts: for the forum and the reading week

1. Week 1 argued that winters punish promises rather than science. If a major AI harm scandal triggered a regulatory backlash, which of this week’s seven stages would you expect to have caused it, and would the backlash be aimed at the right stage?
2. The Robodebt scheme barely used machine learning. Does it belong in an AI ethics syllabus? Argue both sides, then commit.
3. Ireland chose fifteen sectoral regulators rather than one AI regulator. Construct the strongest case for each design, then say which you would choose and what evidence would change your mind.

4. Rudin argues that black boxes should not be used for high-stakes decisions about people. Identify one decision you believe should legally require an interpretable model, and defend it against the strongest objection you can construct.
5. You have read that a faithful model of an unfair world is an accurate model. Where does an engineer's responsibility for that unfairness begin and end? Commit to a position; revisit it in Week 12.

20 Further Reading and Viewing

Everything here is optional enrichment; nothing in this chapter is examinable. Items are ordered by effort, not importance.

Under one hour. The **Gender Shades** project page and its short accompanying video are the fastest route to seeing why disaggregated evaluation matters, and take about fifteen minutes. The original **Datasheets for Datasets** and **Model Cards for Model Reporting** papers are both short, readable and immediately usable as templates for your own project. ProPublica's 2016 **Machine Bias** article, read alongside the vendor's published response, is the best hour you can spend on Chapter 9, precisely because you can watch two competent parties talk past each other.

An evening. The documentary **Coded Bias** (2020) follows the Gender Shades work into policy and is the natural companion to Chapters 4 and 14. Obermeyer and colleagues' 2019 **Science** paper is unusually readable for a journal article and repays reading in full; the mechanism section is the clearest exposition of the proxy problem in the literature. Cynthia Rudin's 2019 argument against explaining black boxes in high-stakes settings is short, forceful, and worth reading even, or especially, if you disagree.

A weekend or more. Cathy O'Neil's **Weapons of Math Destruction** (2016) is the popular book that put this agenda in front of a general audience, organised around scale, opacity and damage, which maps closely onto this booklet's three multipliers. Ruha Benjamin's **Race After Technology** (2019) supplies the sociological frame that engineering accounts tend to lack. Kate Crawford's **Atlas of AI** (2021) follows the material and labour supply chains behind the models. Joy Buolamwini's **Unmasking AI** (2023) is the Gender Shades story from inside. Brian Christian's **The Alignment Problem** (2020), already recommended in Week 1, bridges this week's material to Week 7's RLHF.

Primary papers, for the ambitious. Barocas and Selbst, **Big Data's Disparate Impact** (2016), for the legal framing; Dwork and colleagues (2012) for individual fairness; Hardt, Price and Srebro (2016) for equality of opportunity; Kleinberg, Mullainathan and Raghavan (2016) and Chouldechova (2017) for the impossibility result in its original form; Ribeiro and colleagues (2016) for LIME and Lundberg and Lee (2017) for SHAP; Selbst and colleagues, **Fairness and Abstraction in Sociotechnical Systems** (2019), for the five traps of purely technical fairness work; and Barocas, Hardt and Narayanan's open textbook **Fairness and Machine Learning**, which is the standard scholarly treatment of everything in this booklet and free online.

Appendix: How Week 2 Maps to the Module

Module element	Week 2 connection
LO6 (ethical implications: bias, fairness, explainability)	This entire booklet; assessed in the Week 6 quiz and again in the responsibility analysis of the final project. Moving this material to Week 2 means LO6 is taught and consolidated well before the quiz.
Indicative content 9	Ethical AI: bias, fairness, accountability, transparency. Chapters 2 to 15, with the law in Chapter 14.
Week 1 (history)	Inherited: the ELIZA effect as design responsibility (Ch. 12); MYCIN's deployment gap as the sociotechnical lesson (Ch. 1, 13); Deep Blue's opacity (Ch. 12); benchmark culture as the reason Gender Shades worked (Ch. 4).
Week 3 (training)	The loss function is a values statement; class weights and reweighting are fairness levers before they are tricks (Ch. 7).
Week 4 (model families)	Architecture as inductive bias: sense two of Chapter 2, deliberately chosen and defensible.
Week 5 (science-grounded)	Physics in the loss is the most auditable prior in the module: a constraint you can state, defend and check (Ch. 2, 6).
Week 7 (LLMs)	Training-data provenance and consent (Ch. 4); whose preferences RLHF encodes, as a proxy problem (Ch. 6); test on substance, never fluency (Ch. 12).
Week 8 (scale-free)	Efficiency claims must be measured per group, not only on average (Ch. 7).
Week 9 (lean)	Compression's disparate impact on long-tail cases, and the audit cost of on-device privacy gains (Ch. 7, 15).
Week 10 (systems)	Retrieval and agents move the locus of accountability for a wrong answer; contestability must be designed into agentic systems (Ch. 13).
Week 11 (frontier)	AI for science: whose data, whose access, dual use, and the same seven questions applied to scientific instruments (Ch. 16).
Week 12 (project)	The critical brief carries a responsibility section, marked against Chapter 16's seven questions and the model-card practice of Chapter 15.
Interactive Companion	The Bias & Fairness Explorer is Chapters 7 to 9 as a manipulable object: move the threshold, watch four

	metrics move in different directions, and discover the impossibility result by hand.
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End of the Week 2 Reference Booklet. Next: Week 3, how machines learn, fast, where the darts loop of 1958 becomes gradient descent, backpropagation, and a loss function you now know is a statement of values.